Box Robot Base with Attachment Points and Motors

OMMS FLL Team # 1 45775

version 1.0, rev 7   29 December 2025

This is a classific box robot design intended to provide two forward facing attachments or a combination of forward and sidefacing attachments. We usually configure the left attachment motor for our worm drive attachment and the right attachment motor, rotated at 90 degrees, is usually for a geared lifting arm. We also use the right motor to drive a forward facing axle to turn objects on the FLL board (like the gears in Masterpiece Mission 12 rotating the chicken or Unearthed Mission 11/Angler Aritfact).

We recommend using four large Spike Prime motors. If you only have two large motors use them for the attachments and use the medium Spike Prime motors for the wheels. Using the medium motors will require adjustments to the design of the front assembly section.

When assembling the robot care needs to be taken on how the sensor and motor cables need to be routed to the hub prior to joining the front and rear secitons. Secure the cables using multiple wire connectors (49283) in the rear compartment containing the hub.

It is loosely based on:

https://www.youtube.com/watch?v=PCMEvf4p2PI&t=2s

The left attachment motor is intended to be used with our Wormscrew Attachment.

The right attachment motor is intended to be used with our Multipurpose Lifting Arm Attachment.

Section 1 of 2

Front Assembly

Be careful on how attachment motors are placed on top of the yellow plate covering the driving motors.

The primary mechanism the drive motors are secured is via attachment to this plate.

You need to optimize the number of pins (8) from the drive and attachment motors that are connected to the top and bottom of the plates.

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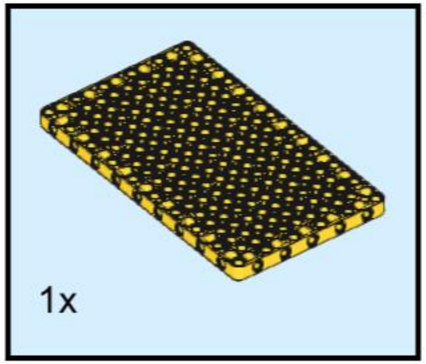

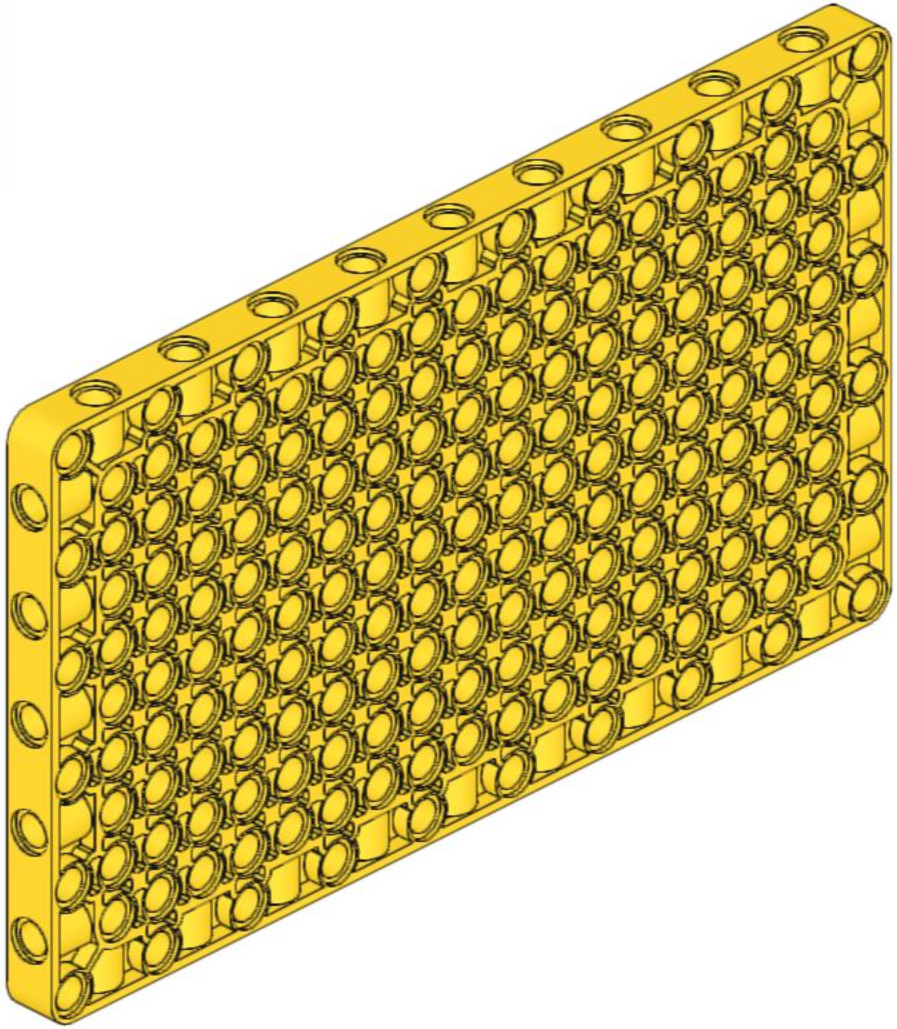

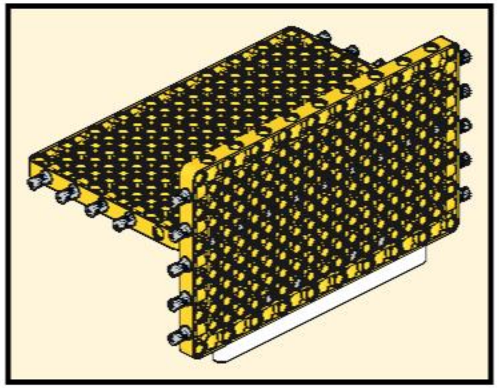

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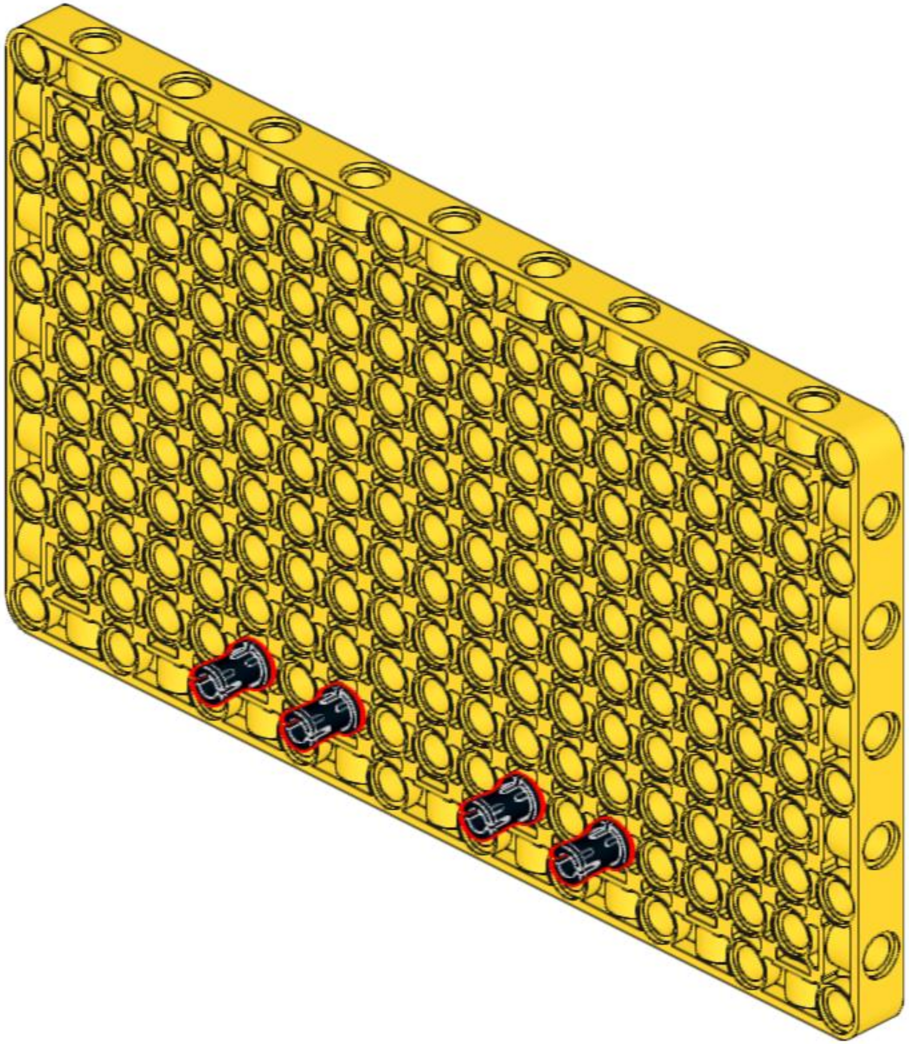

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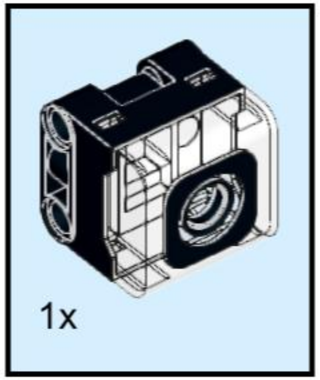

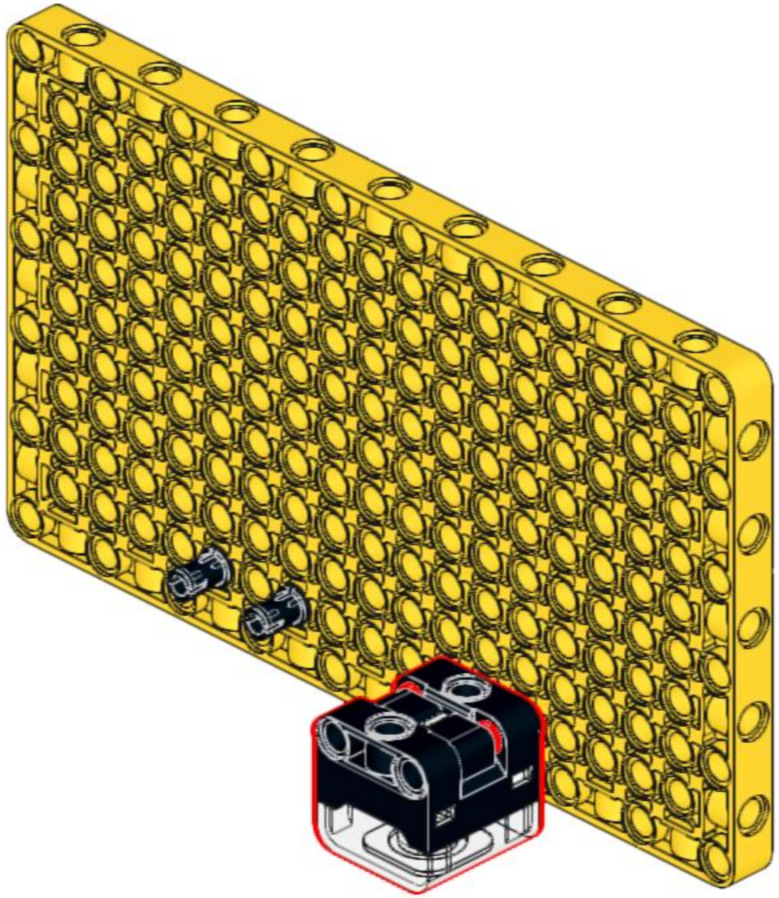

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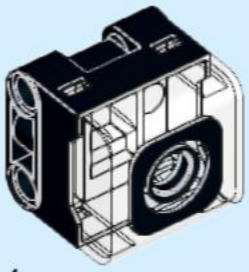
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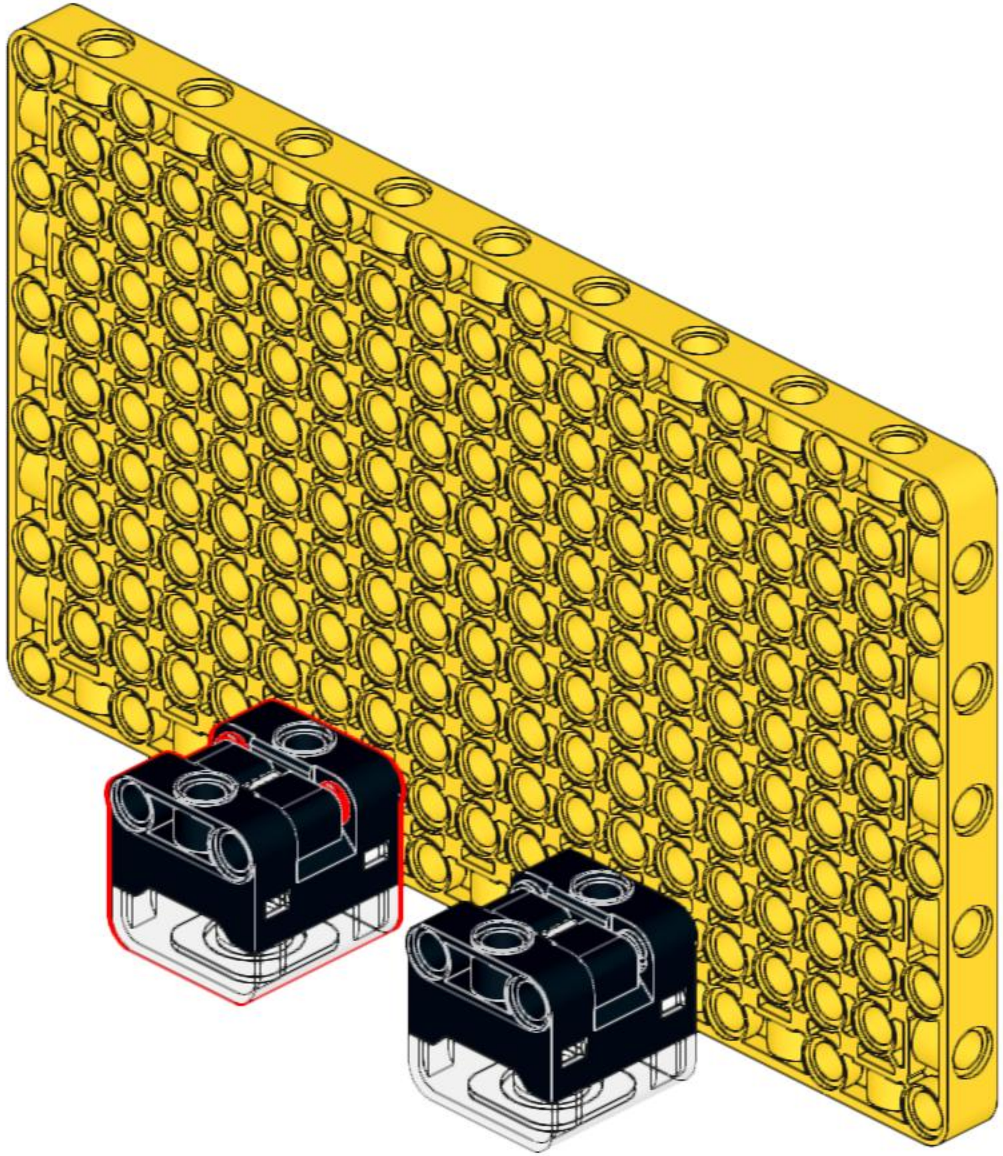

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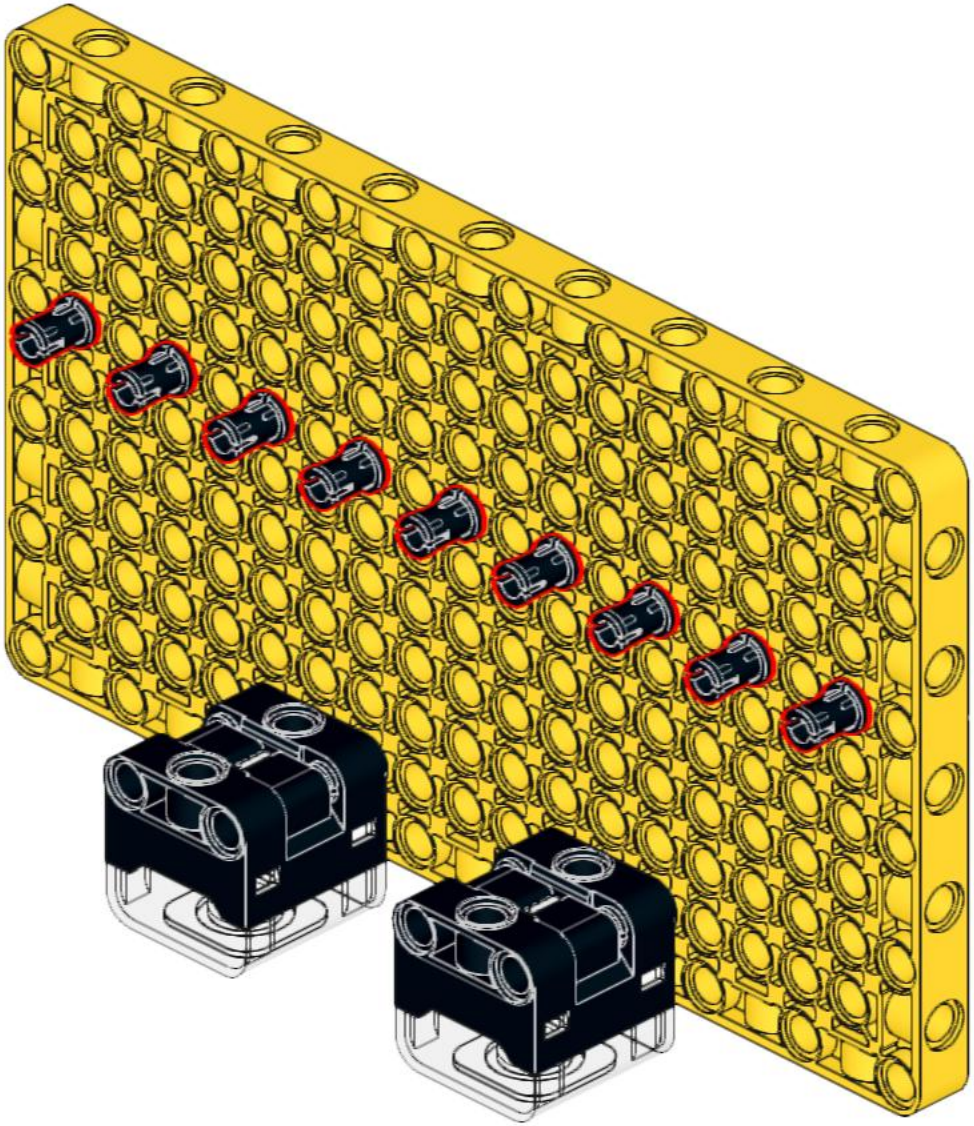

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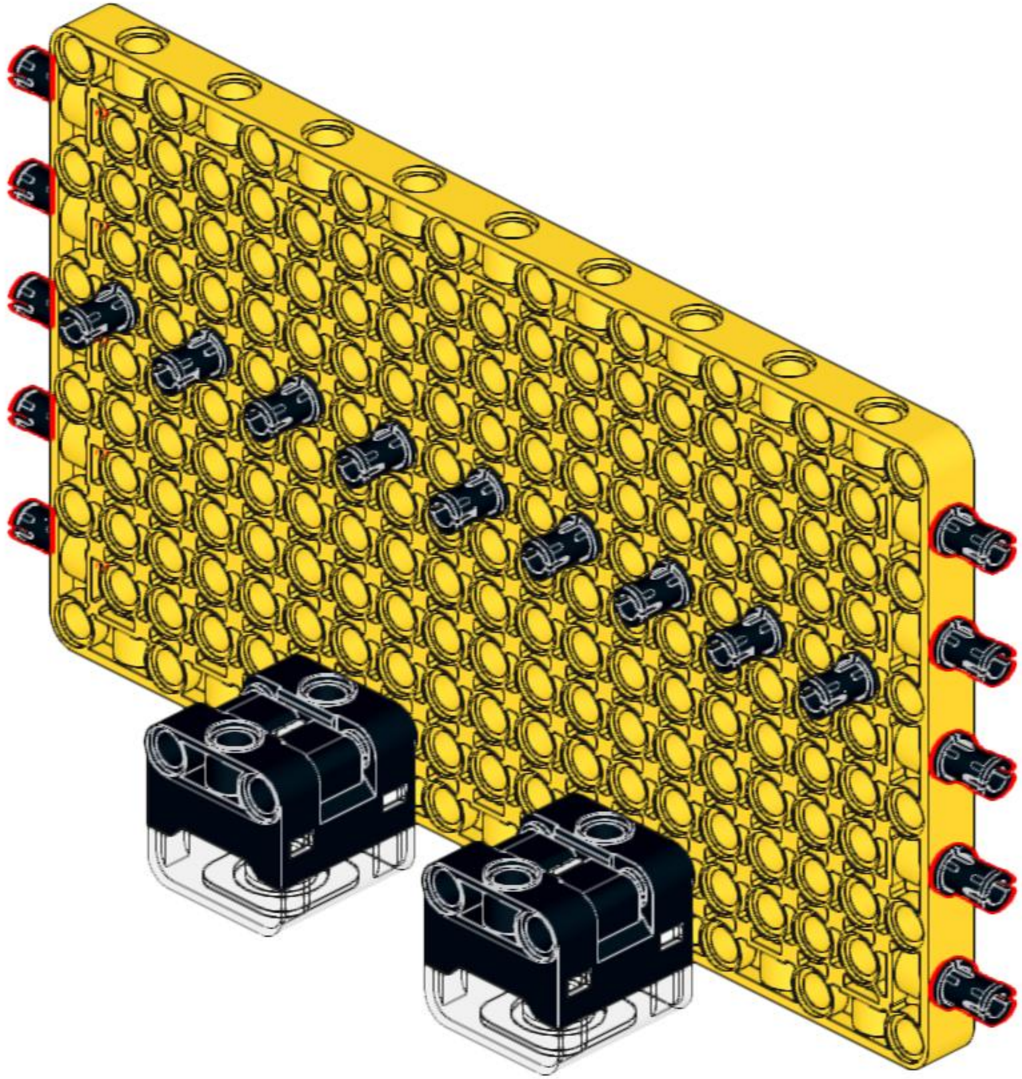

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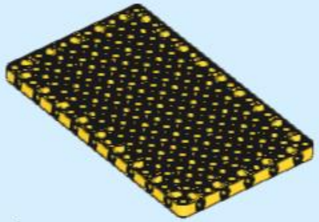
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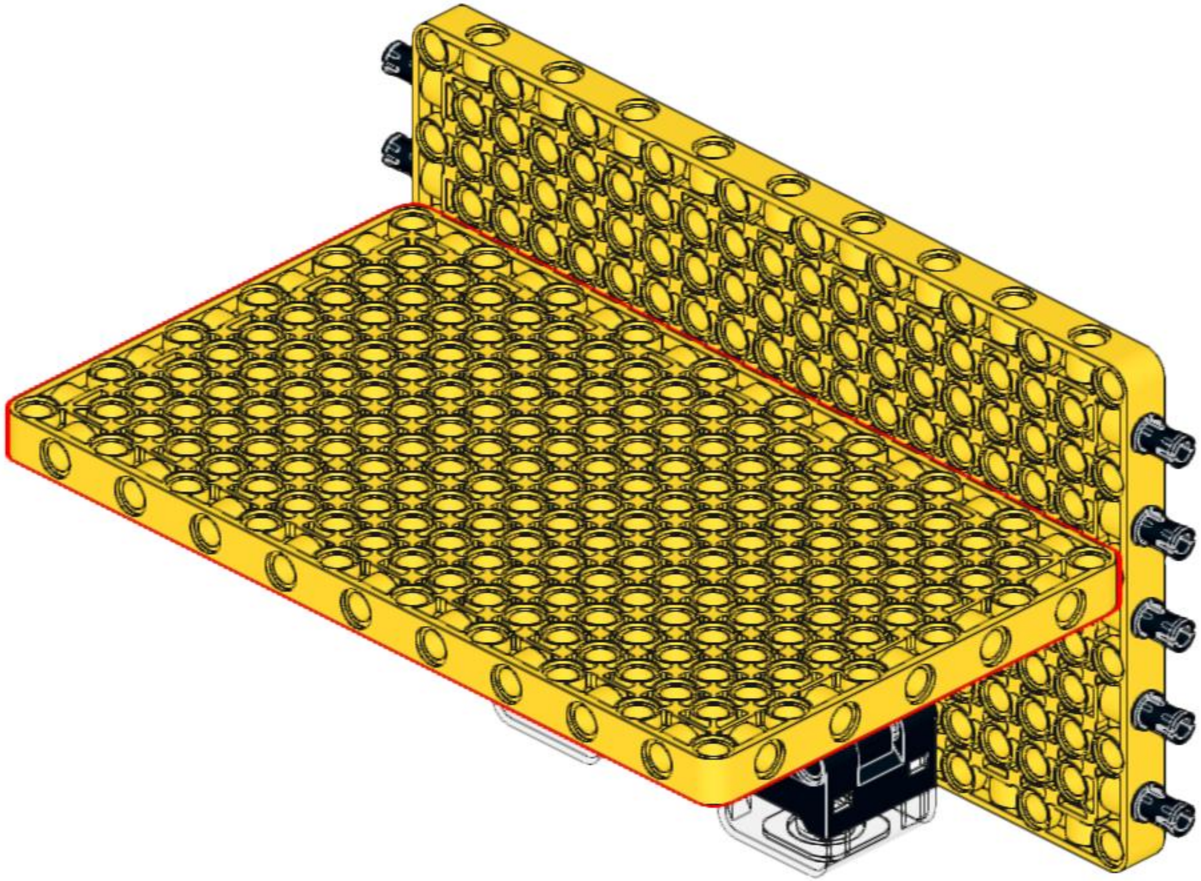

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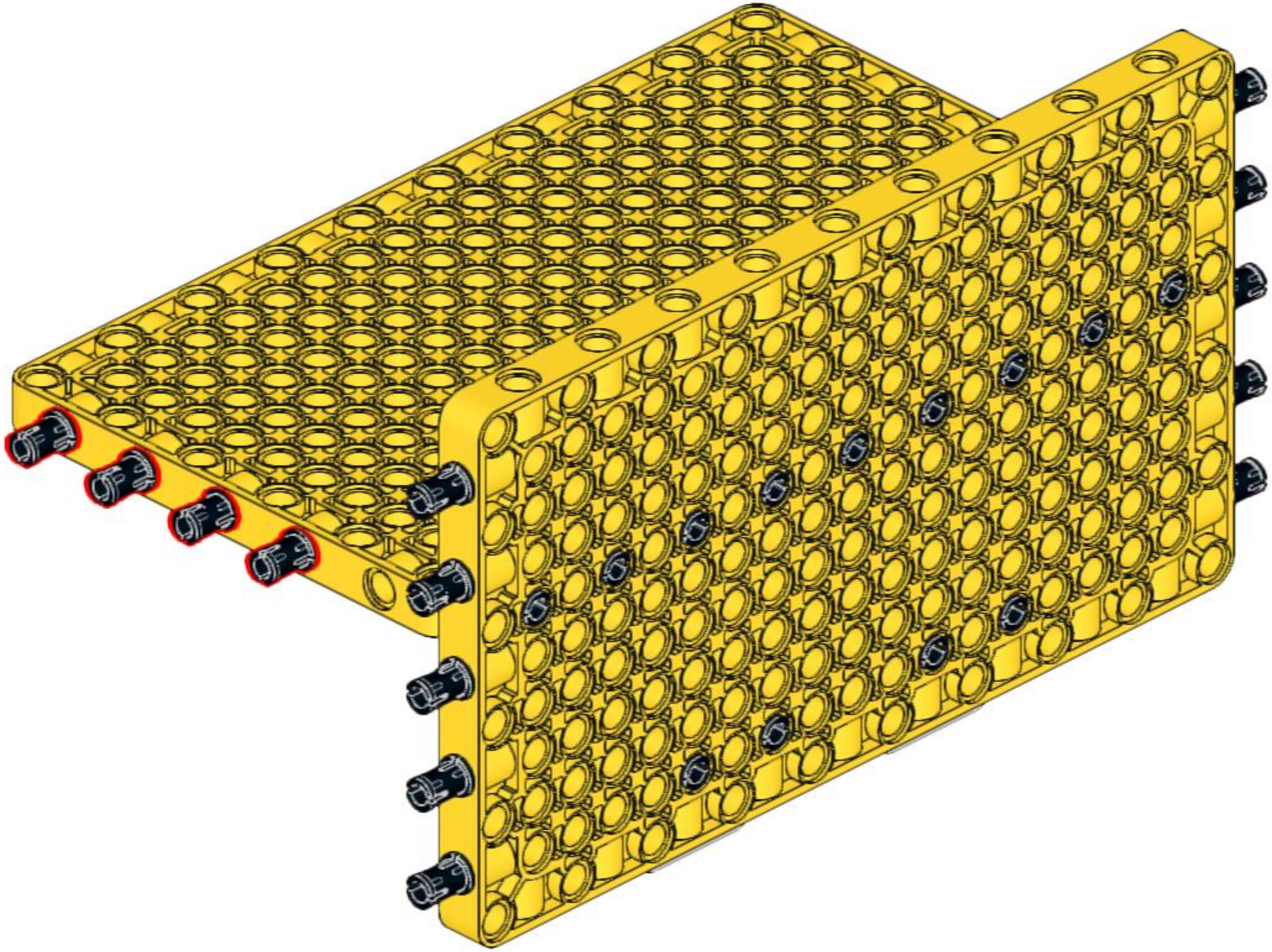

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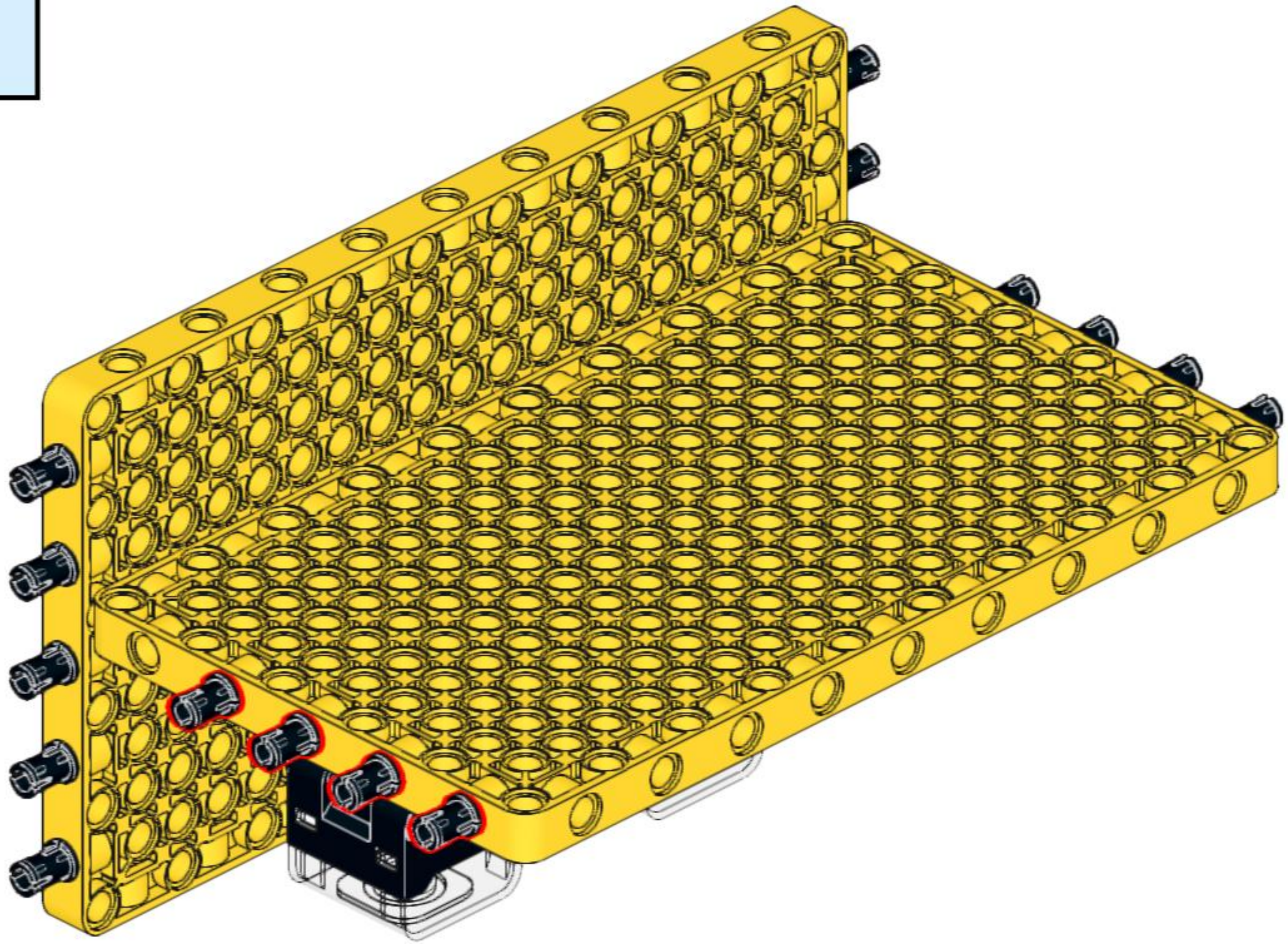

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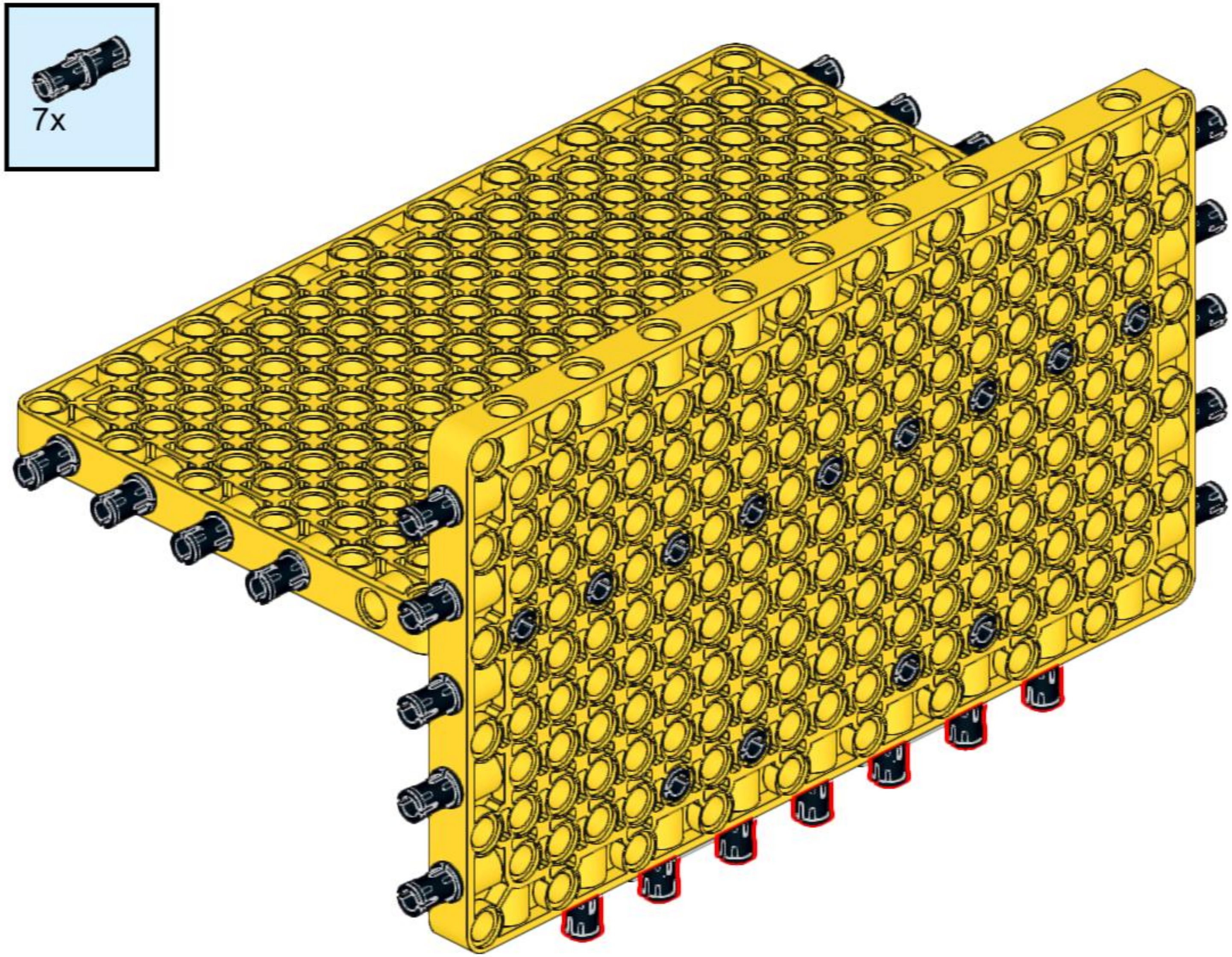

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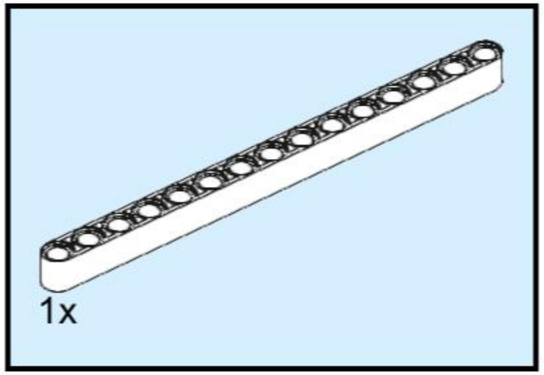

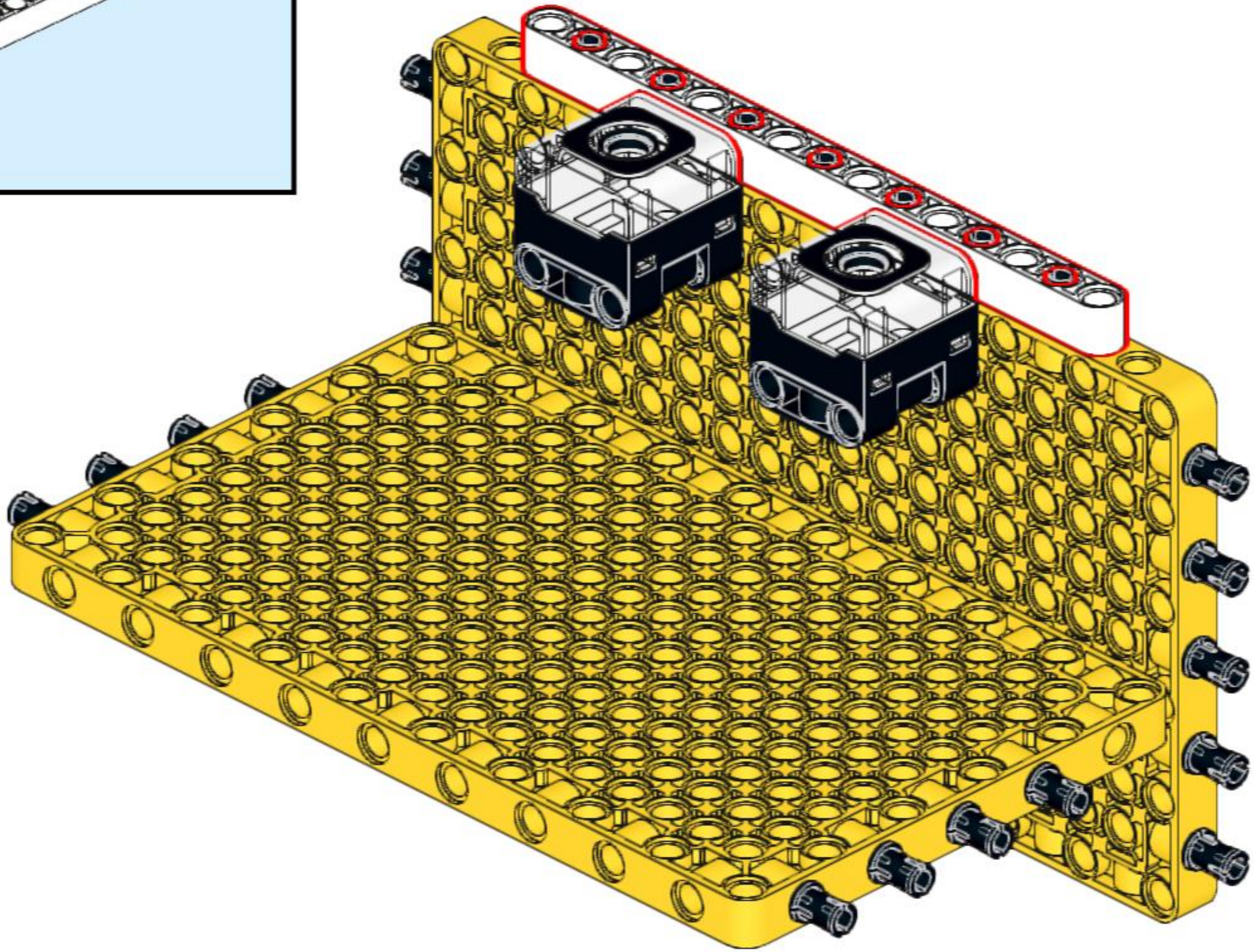

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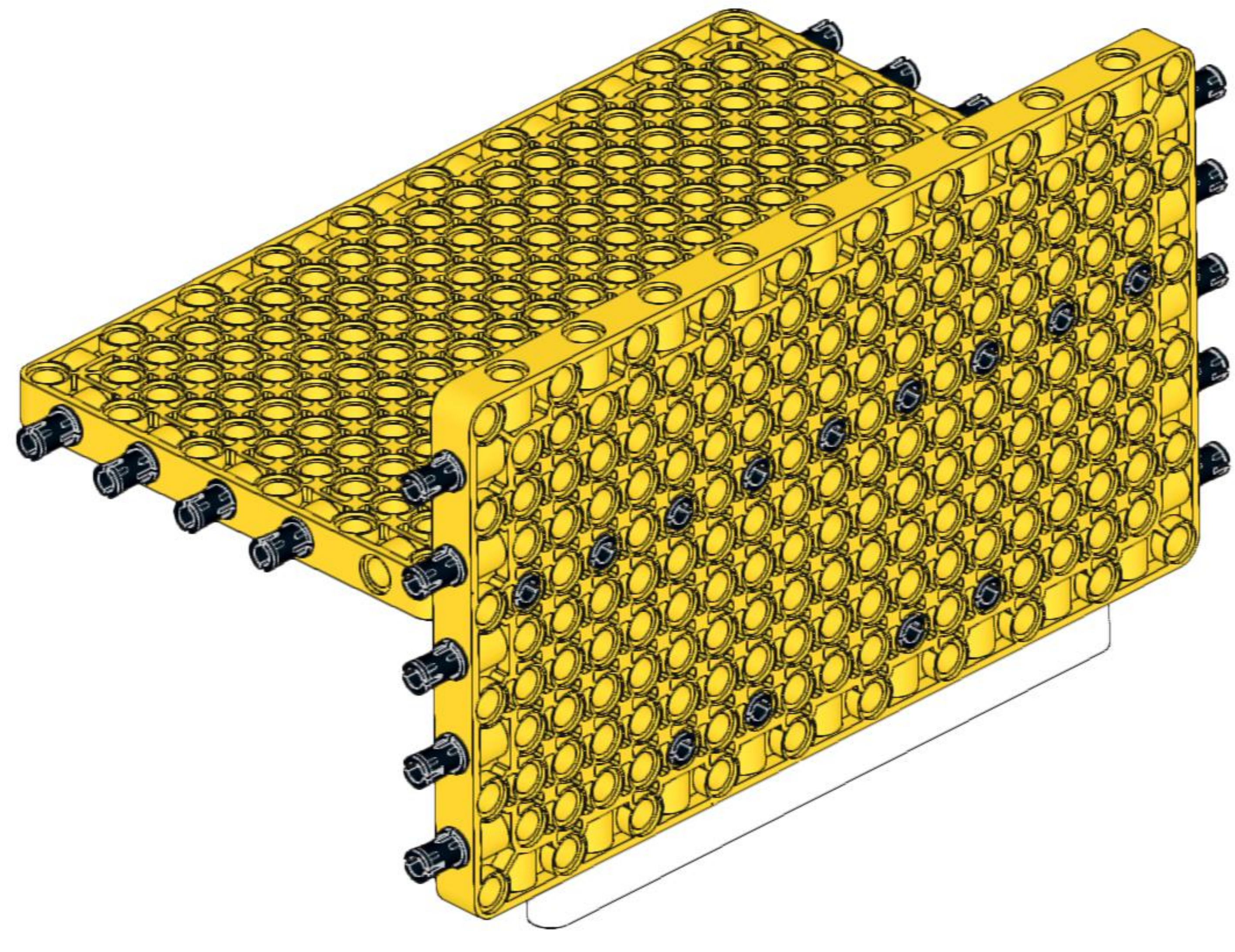

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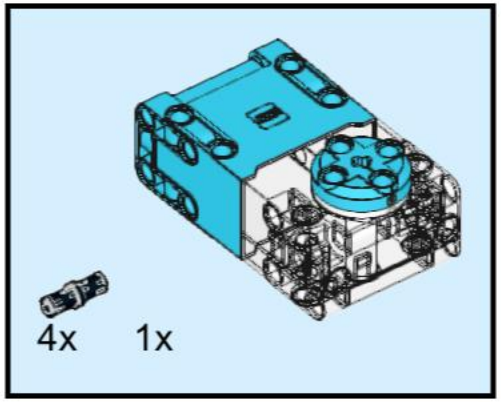

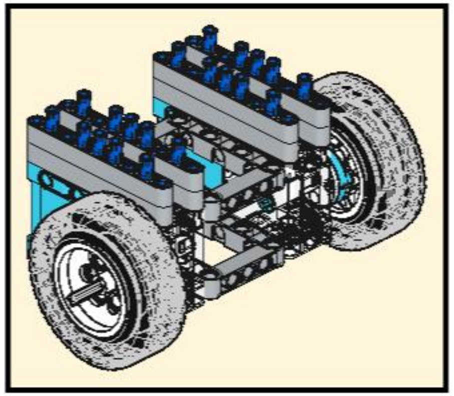

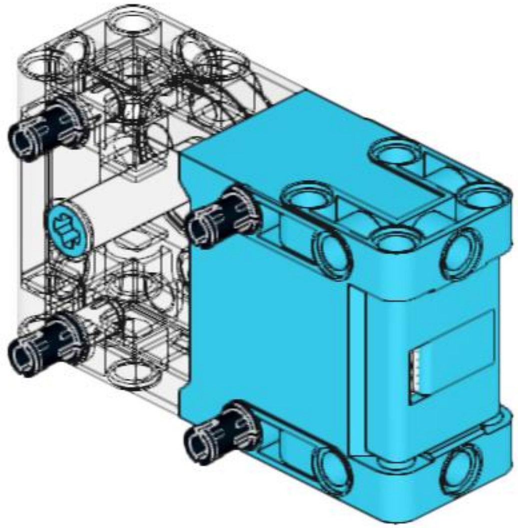

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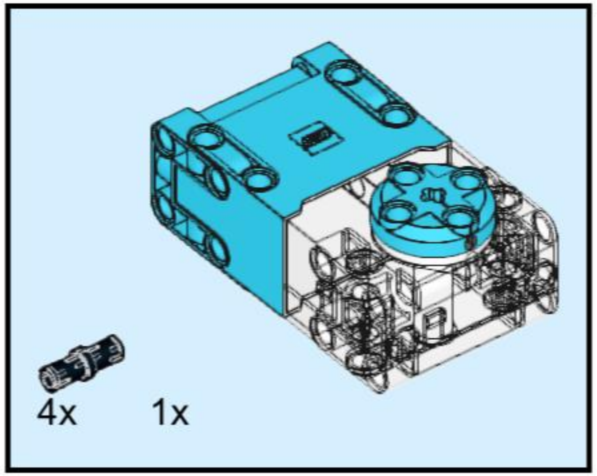

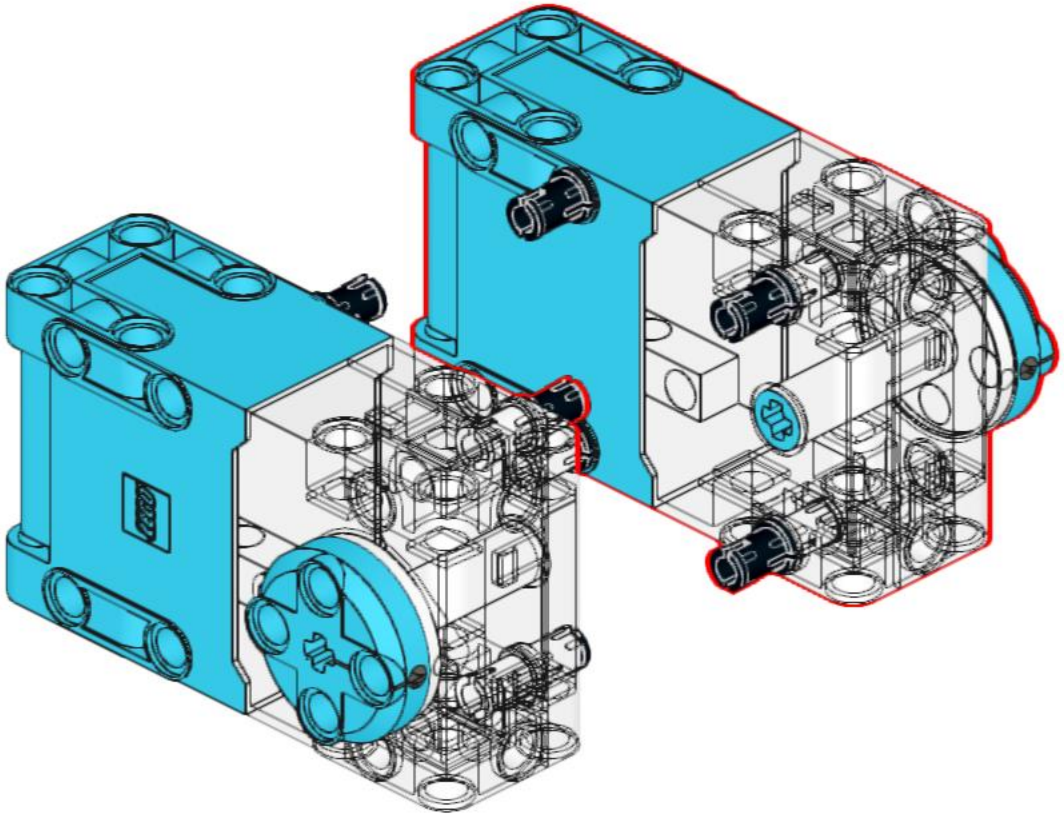

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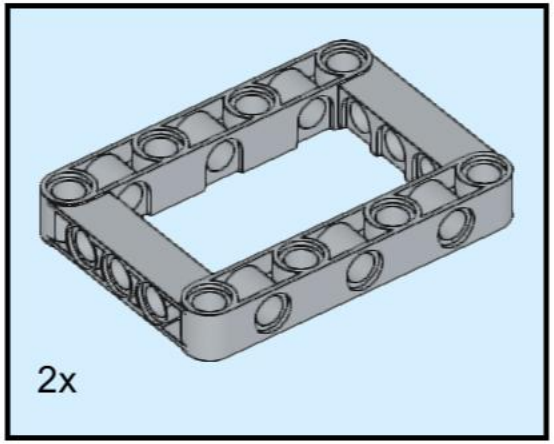

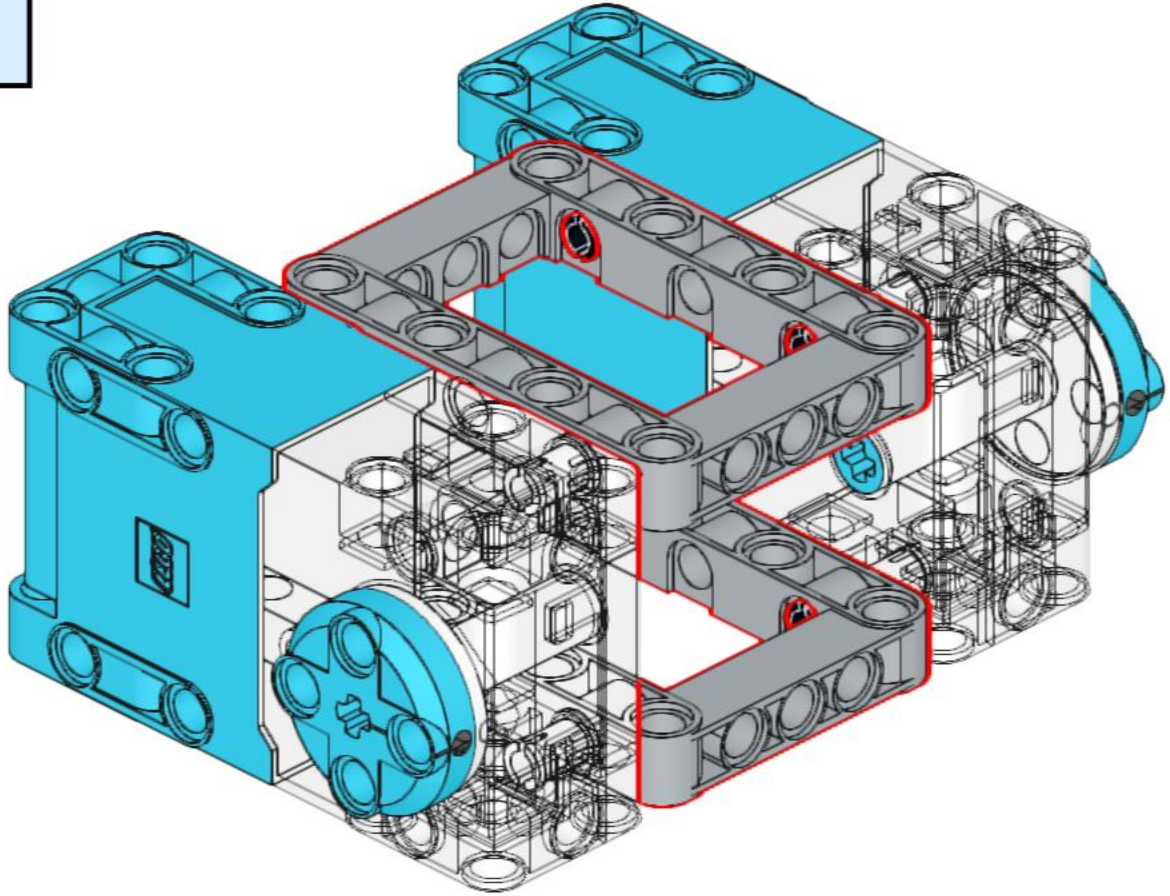

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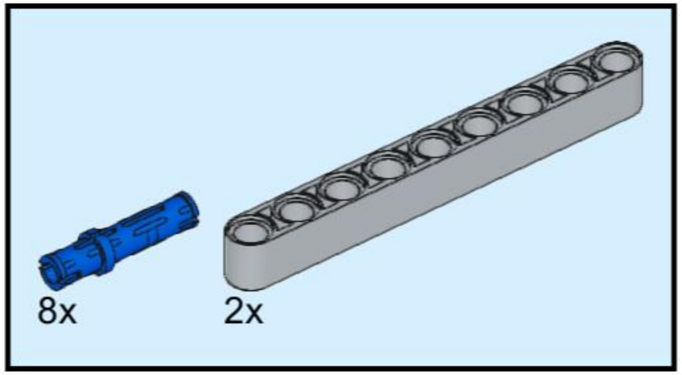

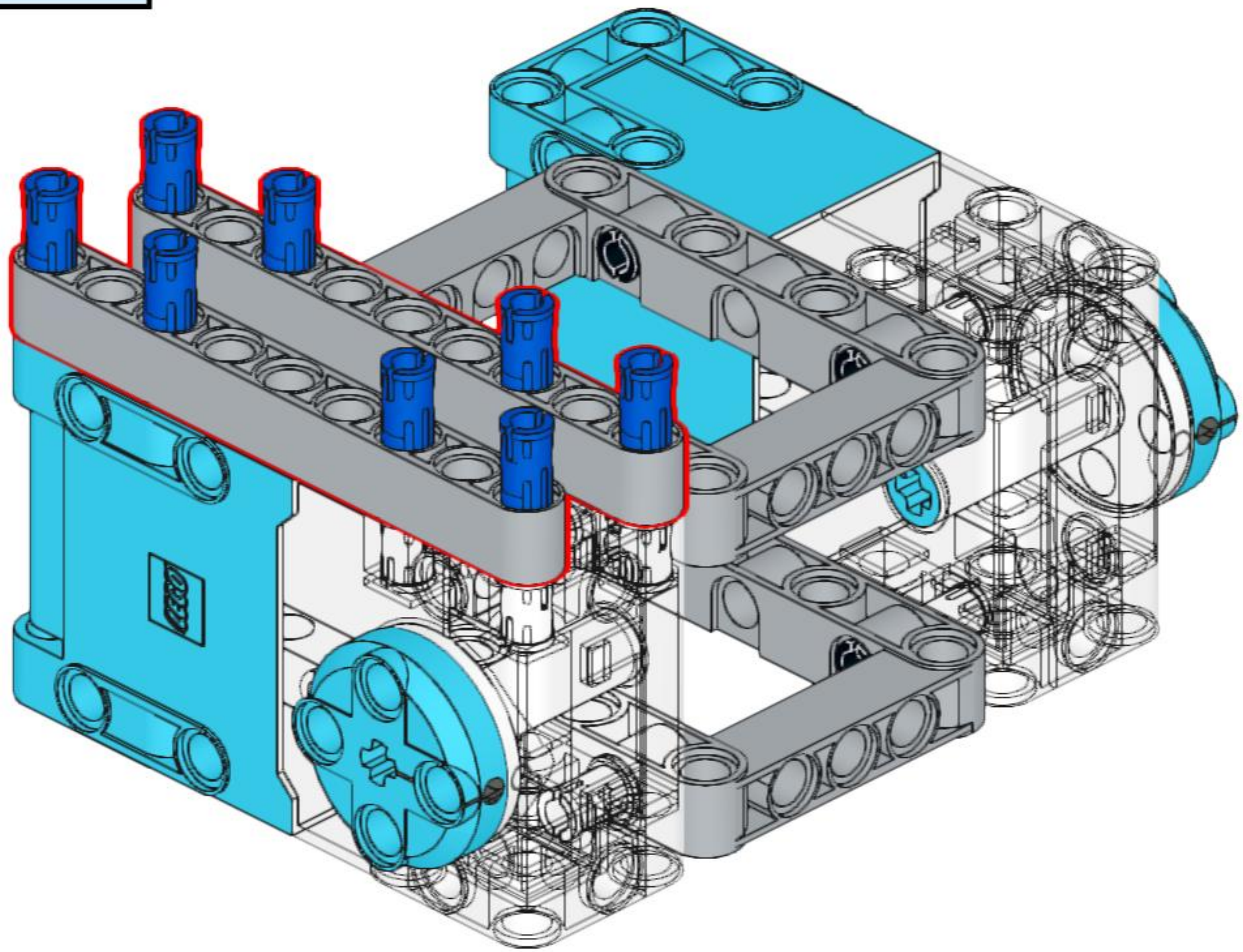

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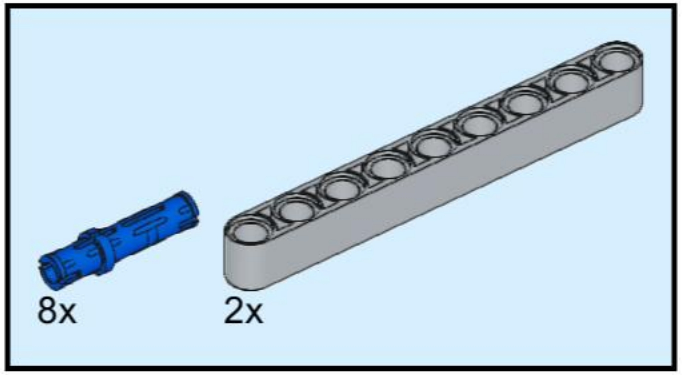

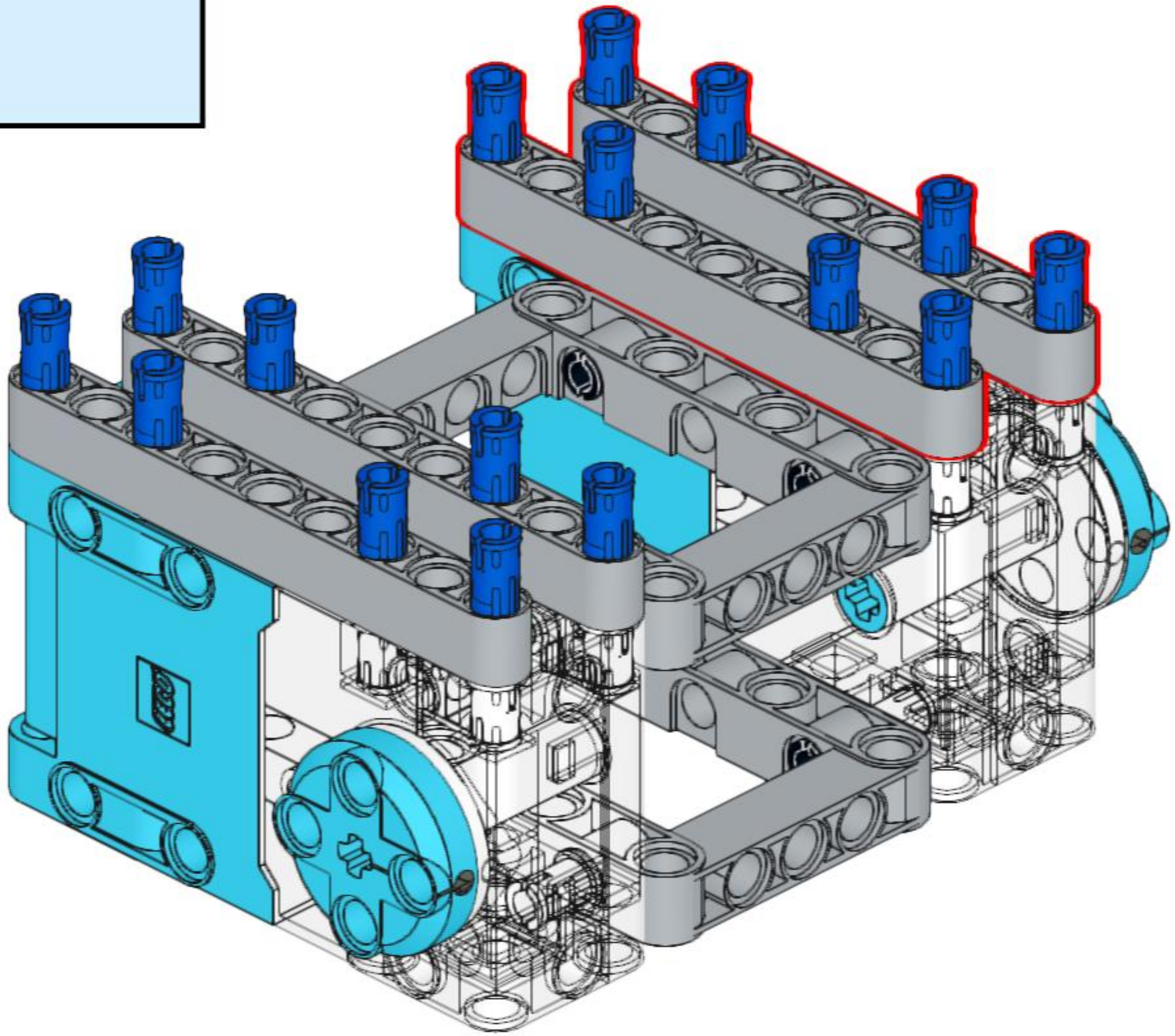

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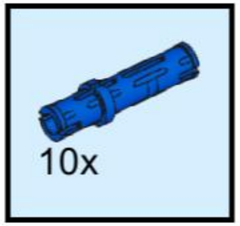

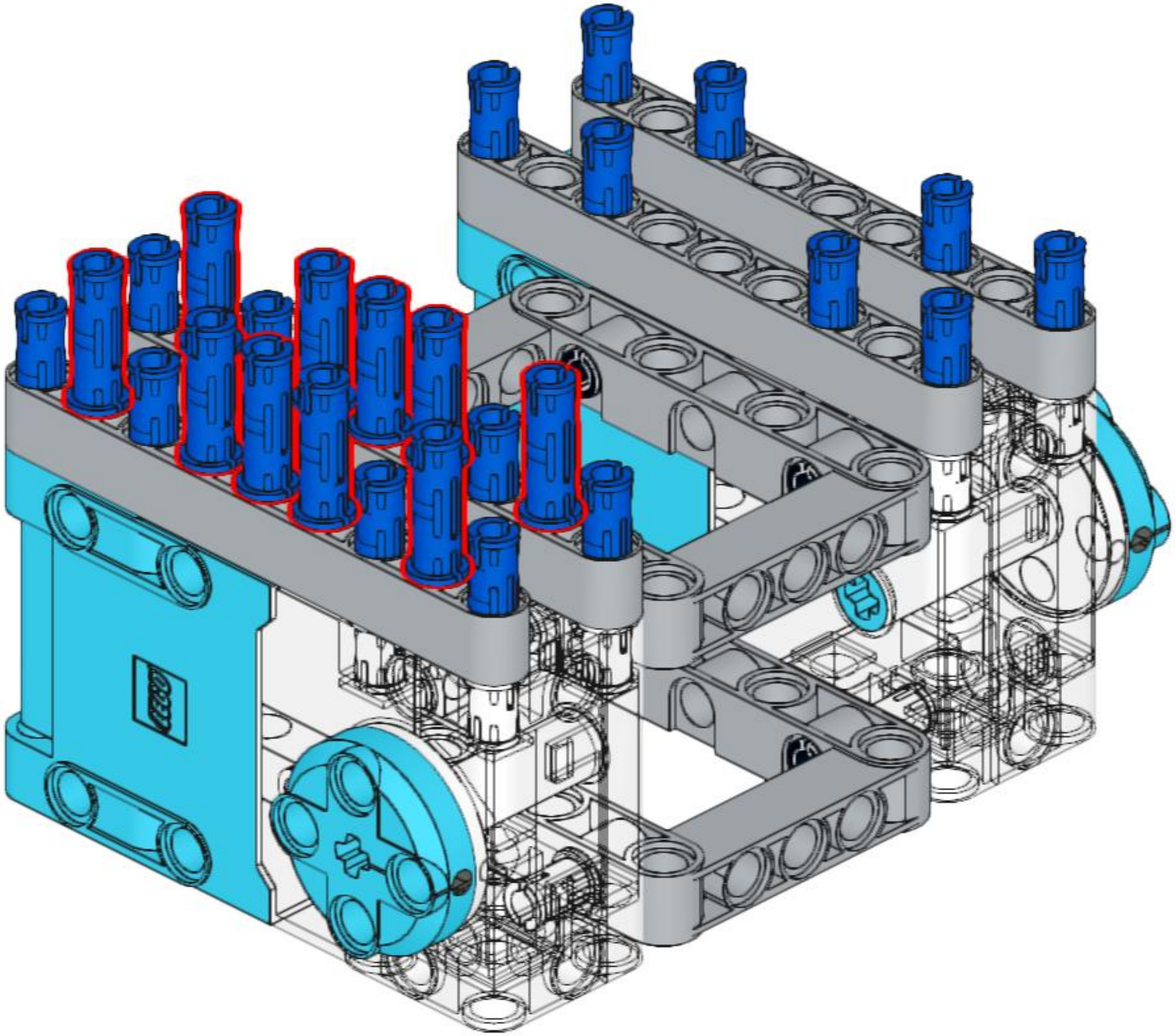

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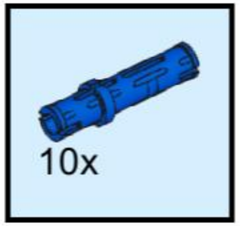

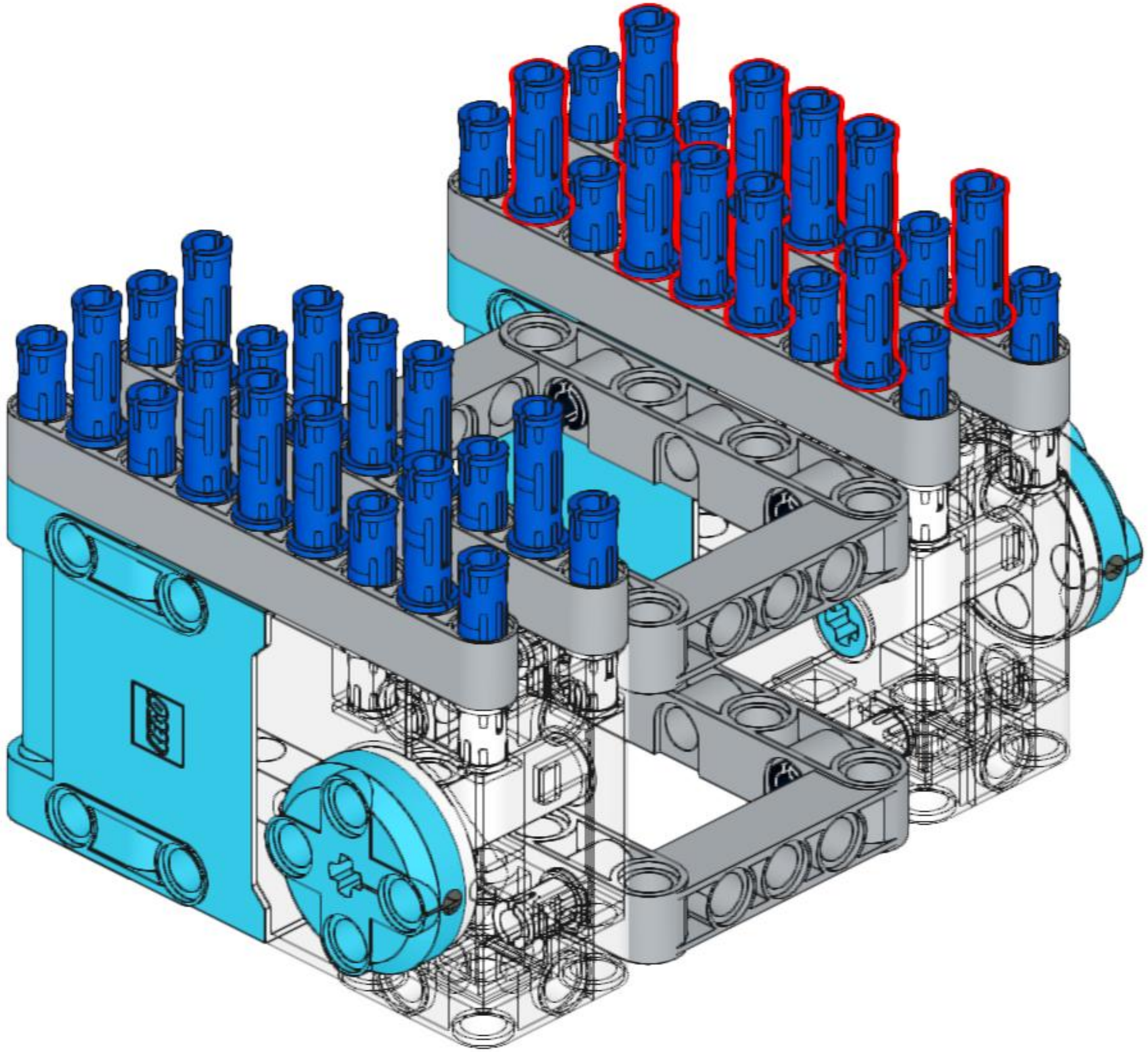

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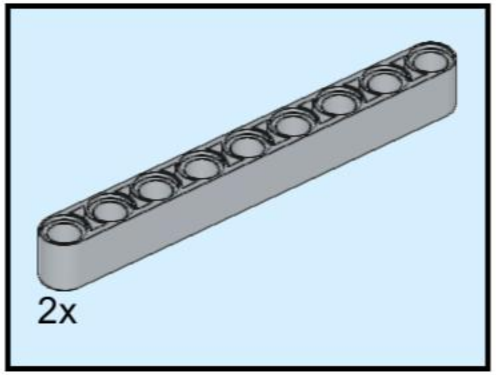

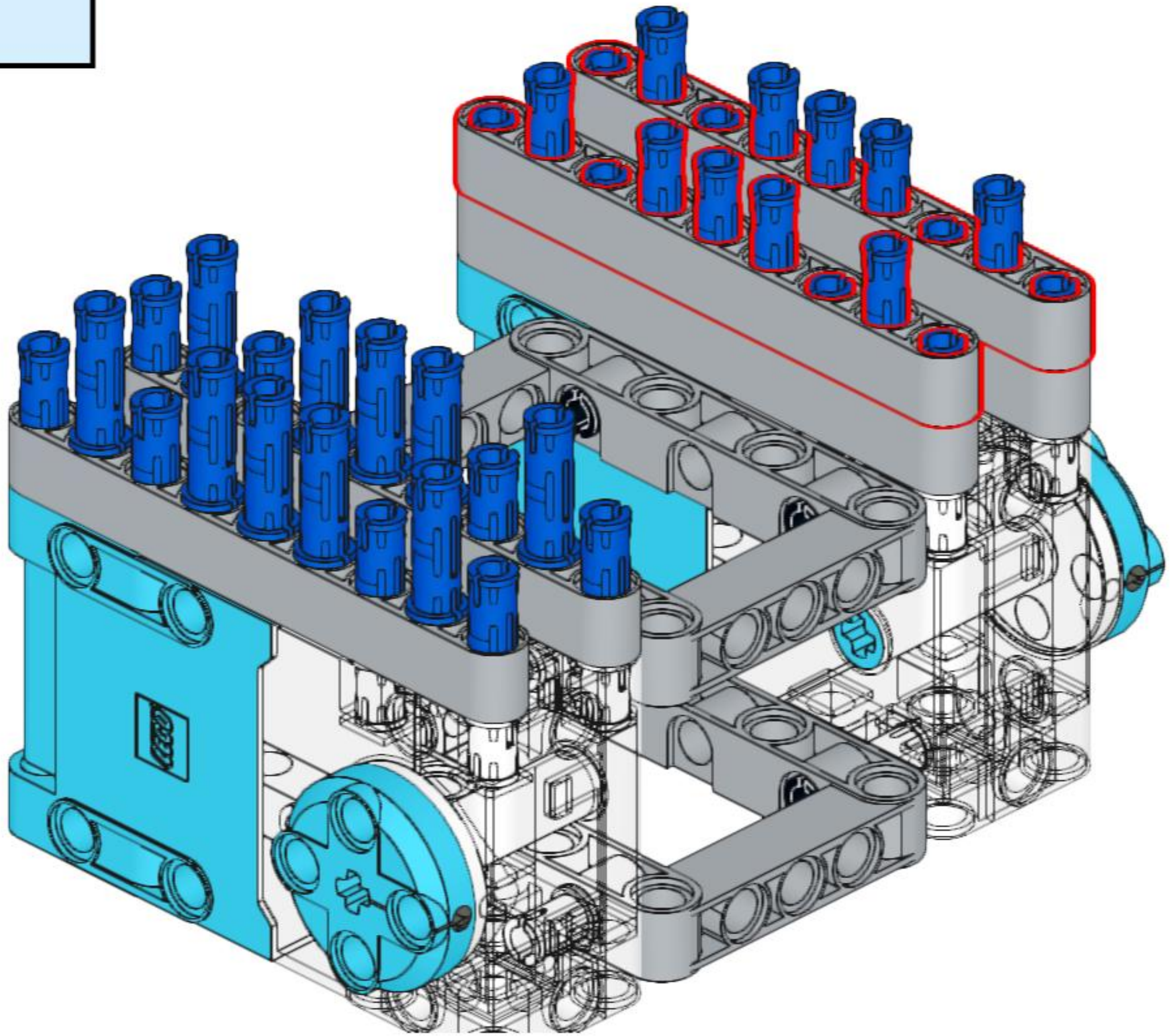

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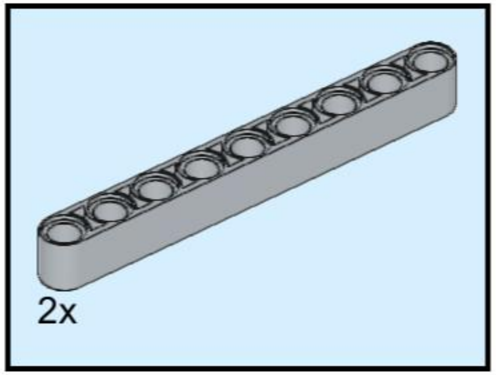

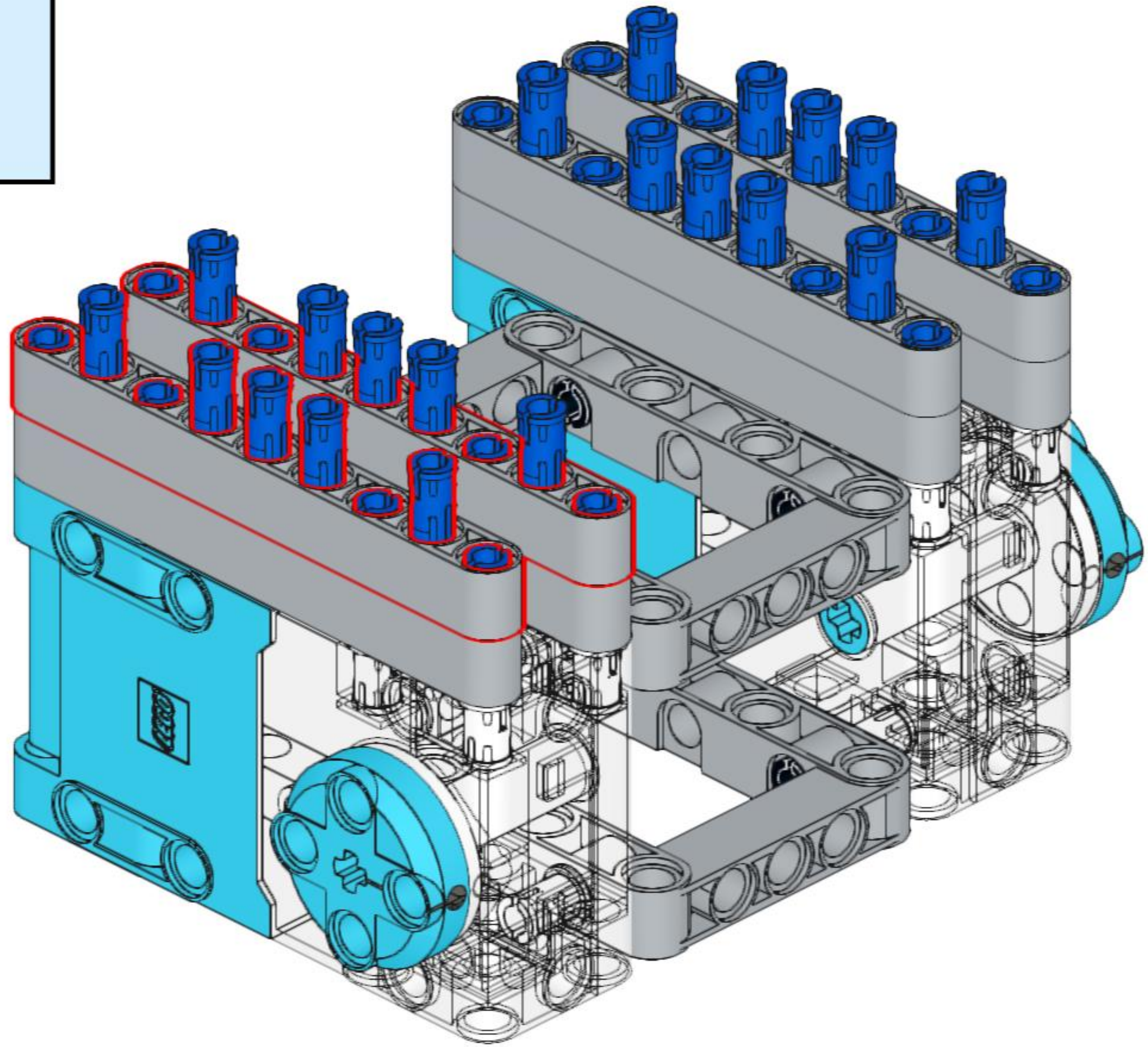

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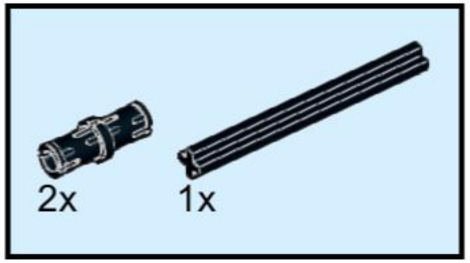

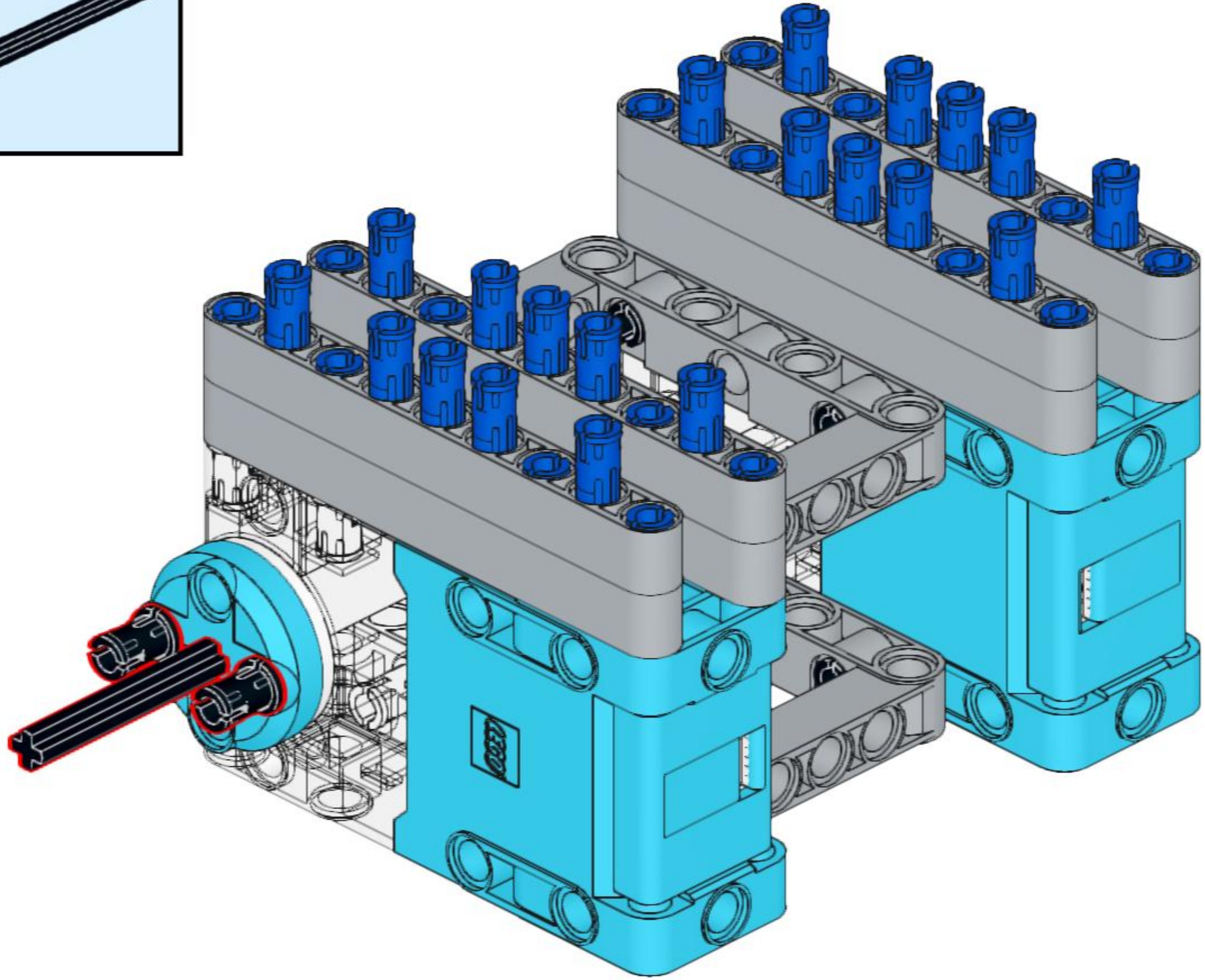

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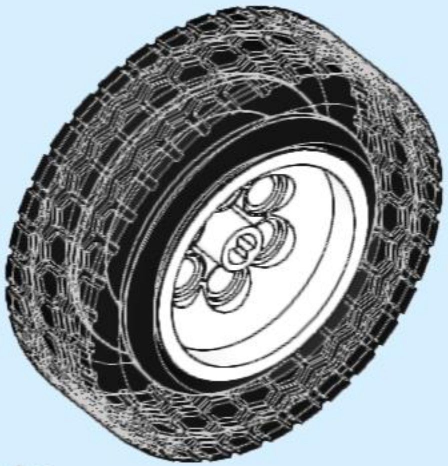
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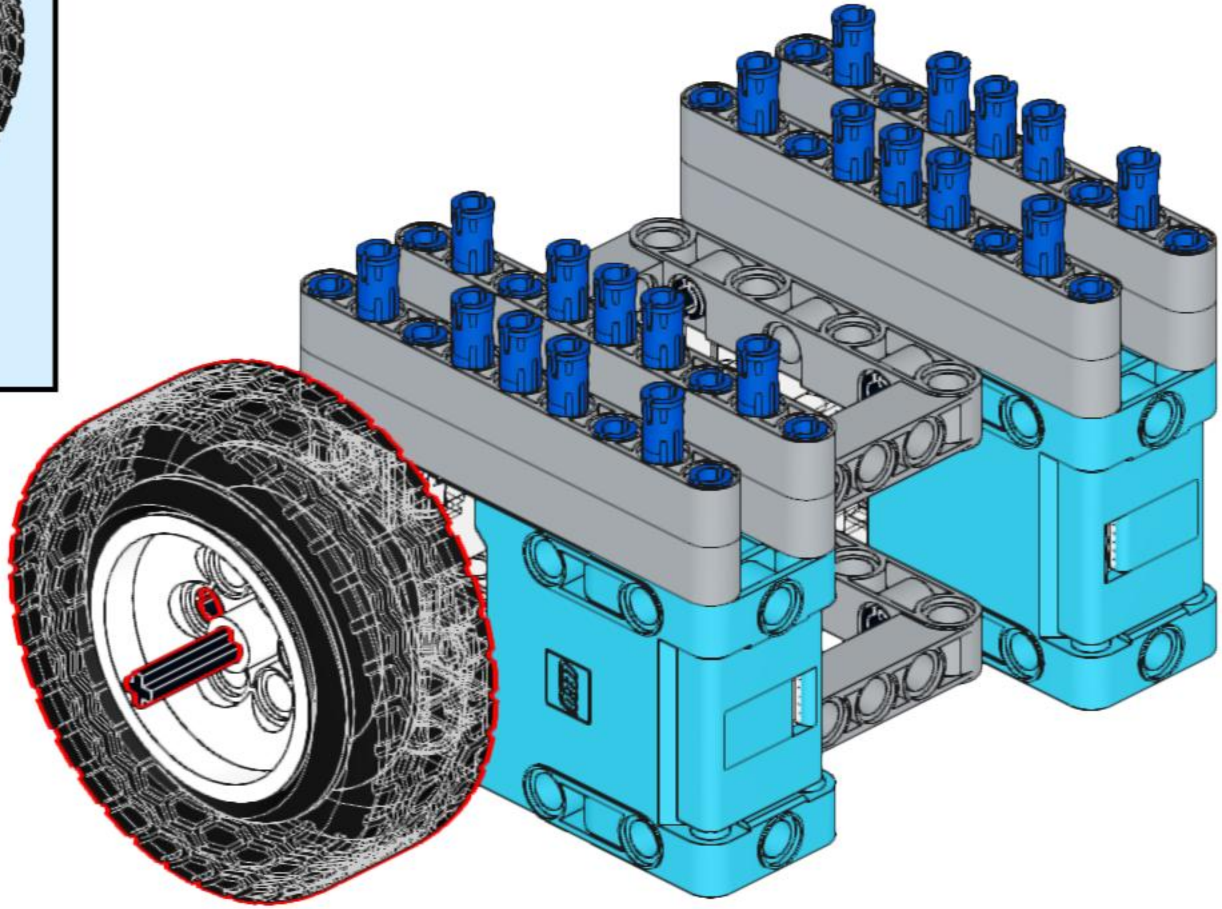

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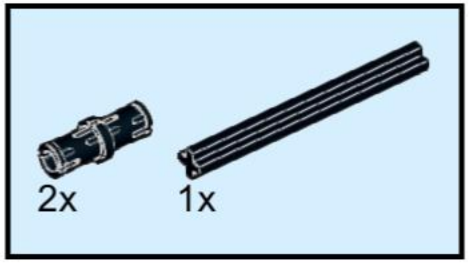

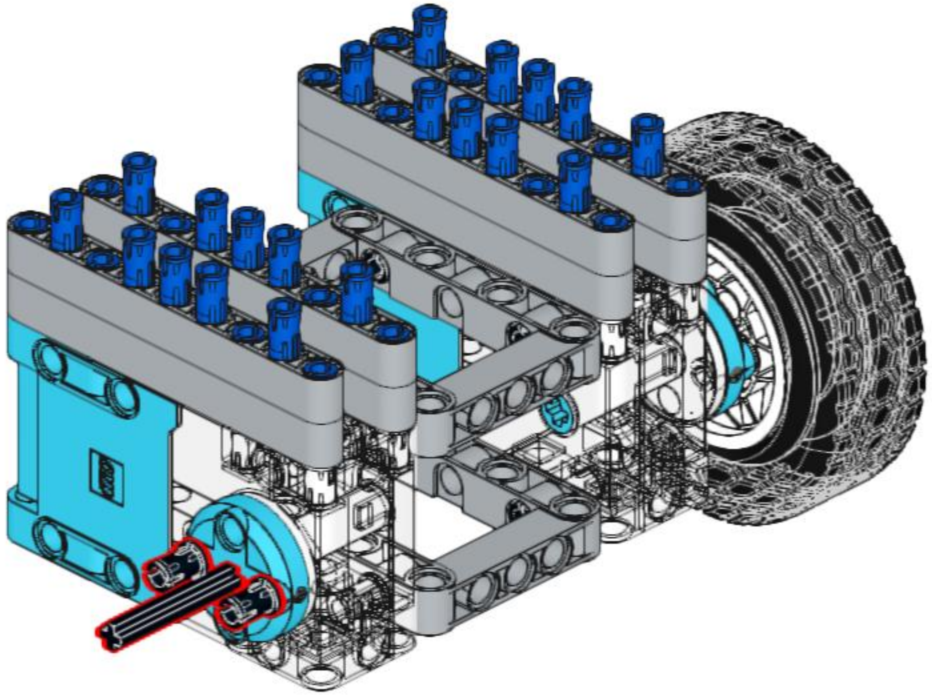

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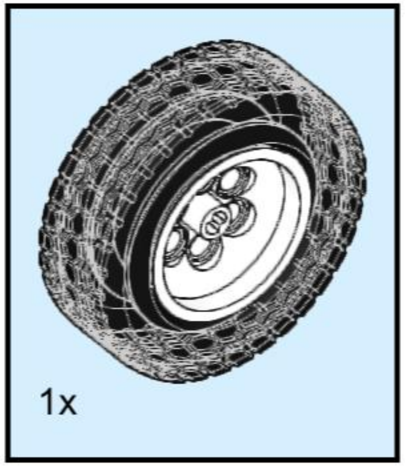

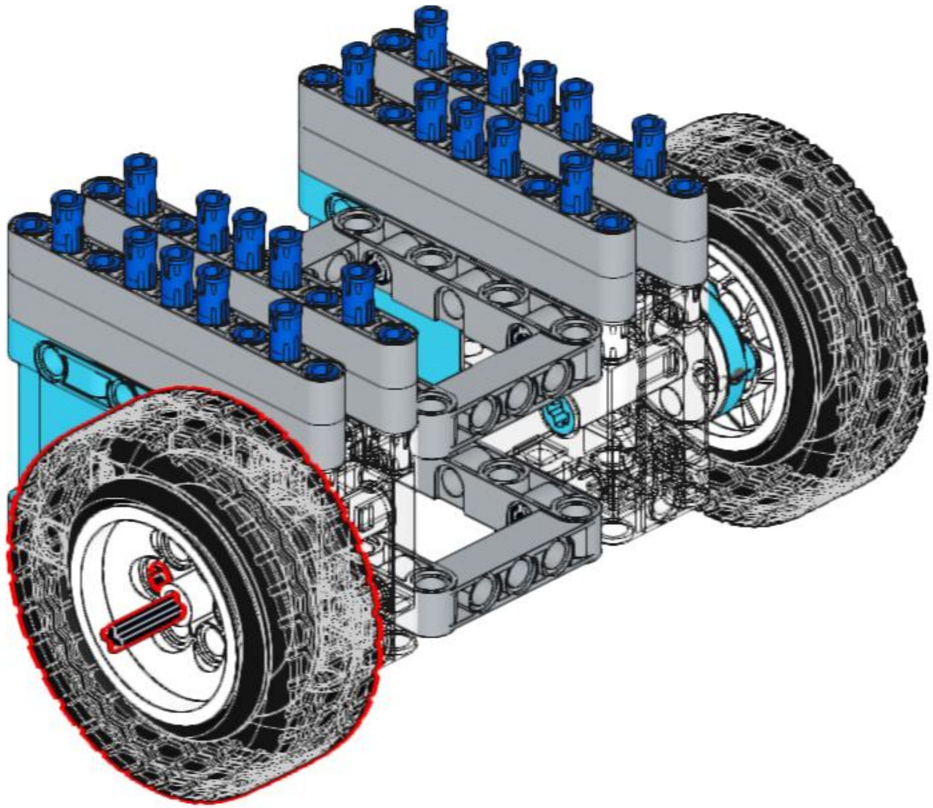

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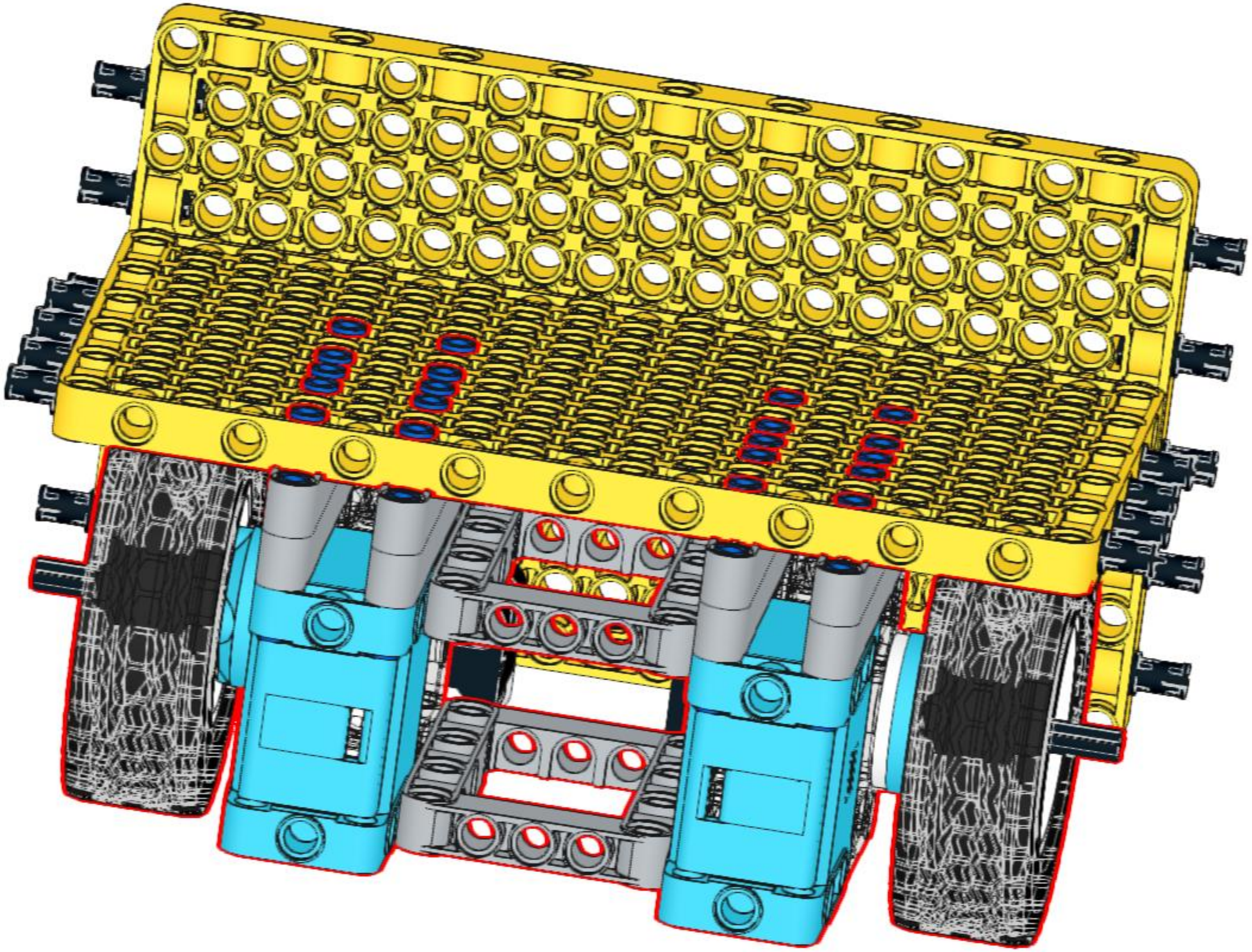

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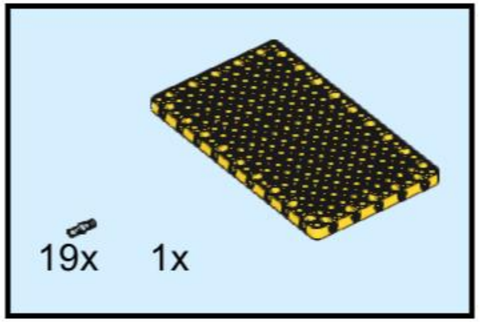

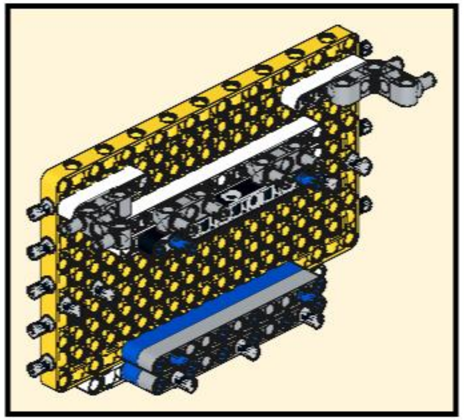

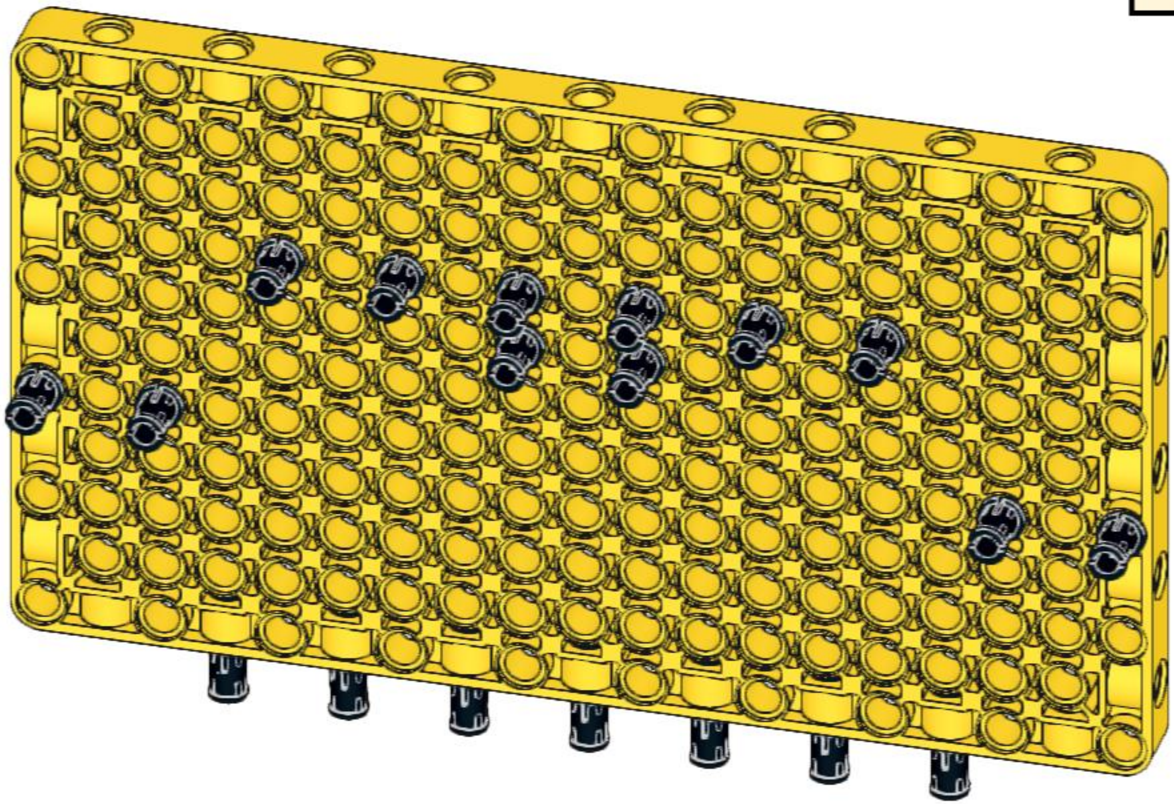

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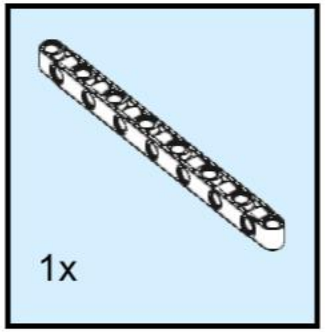

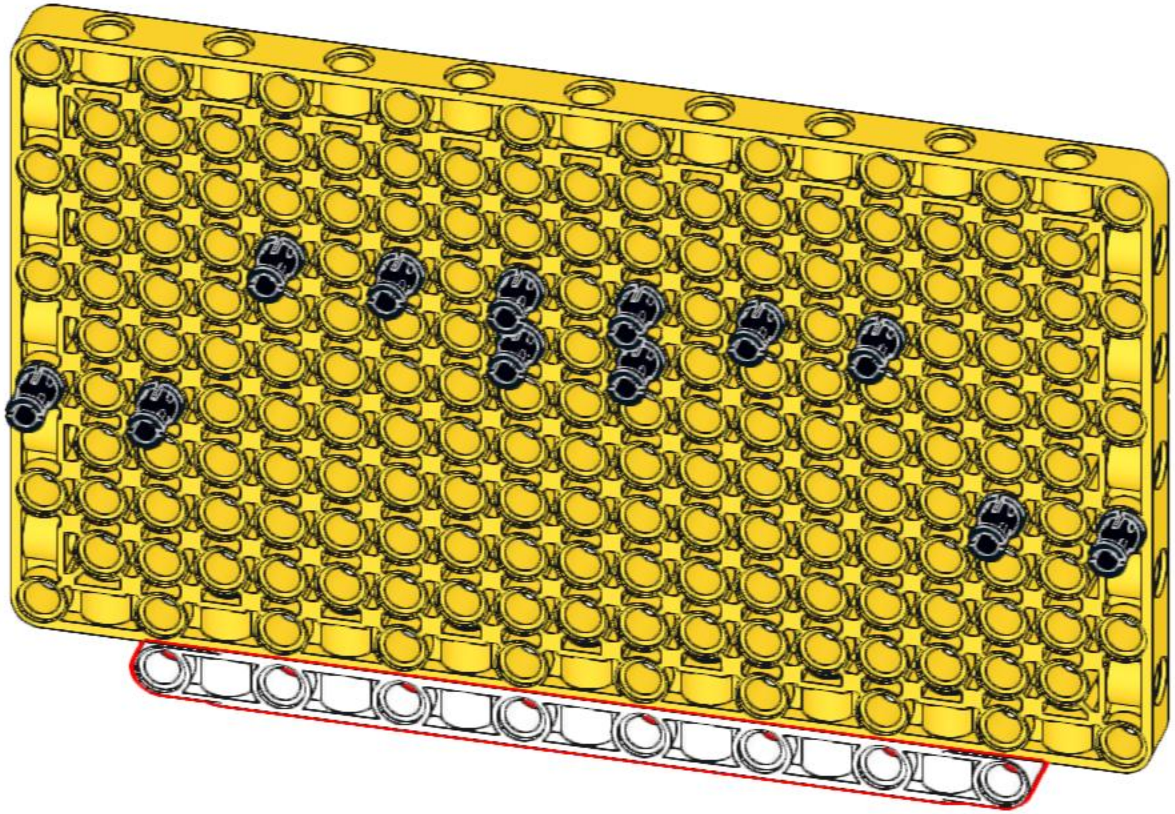

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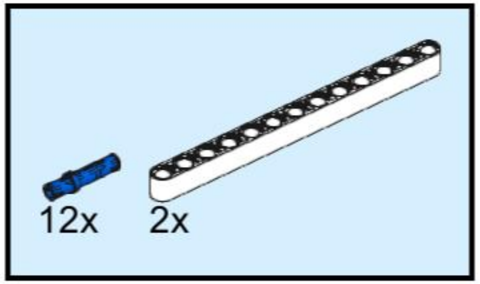

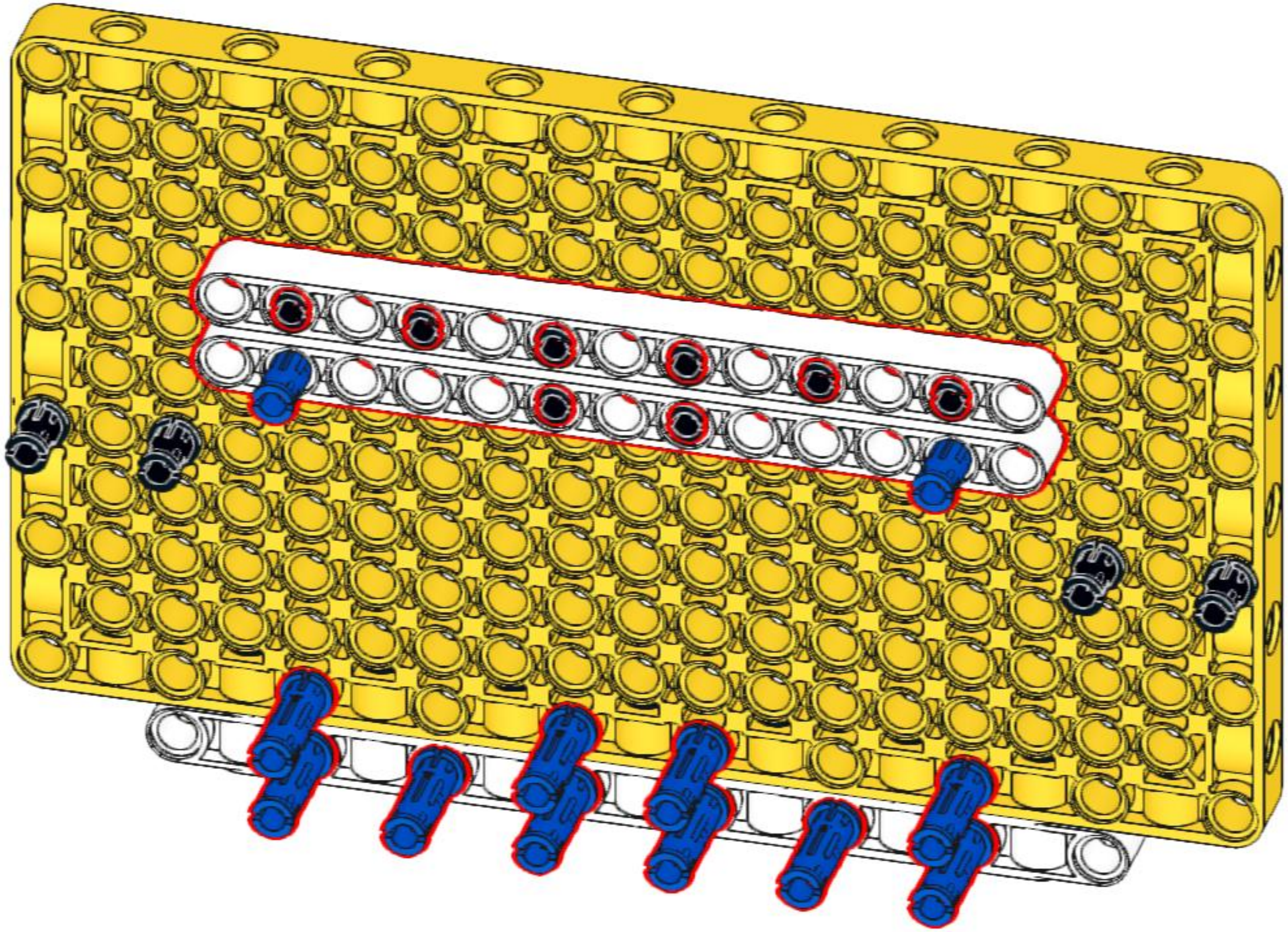

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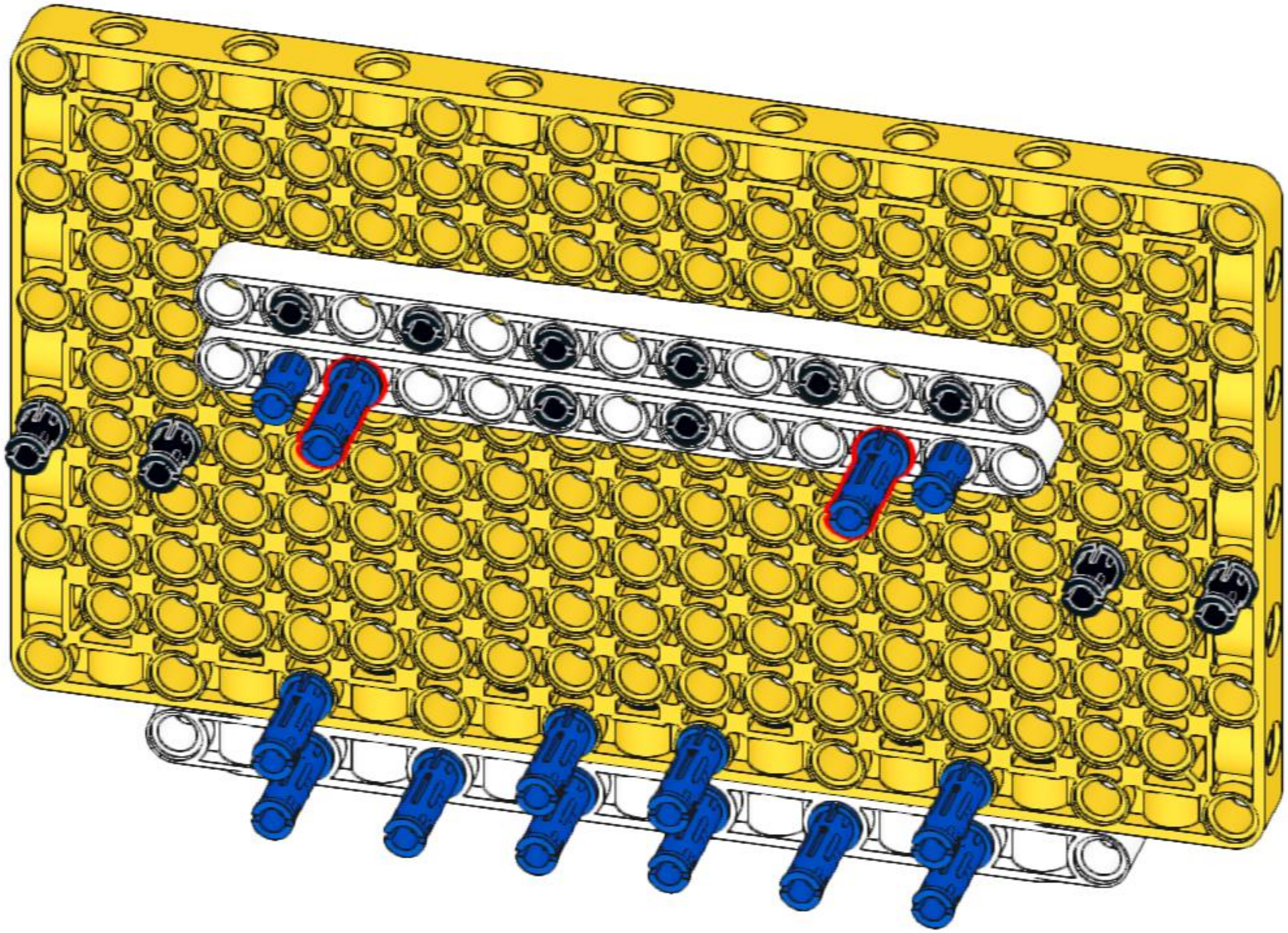

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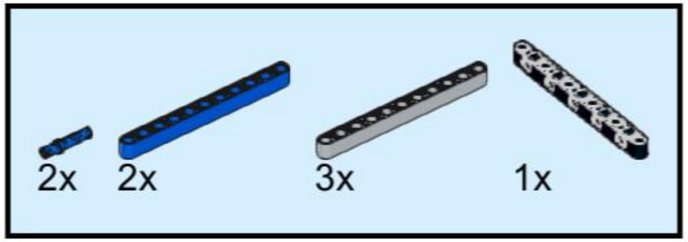

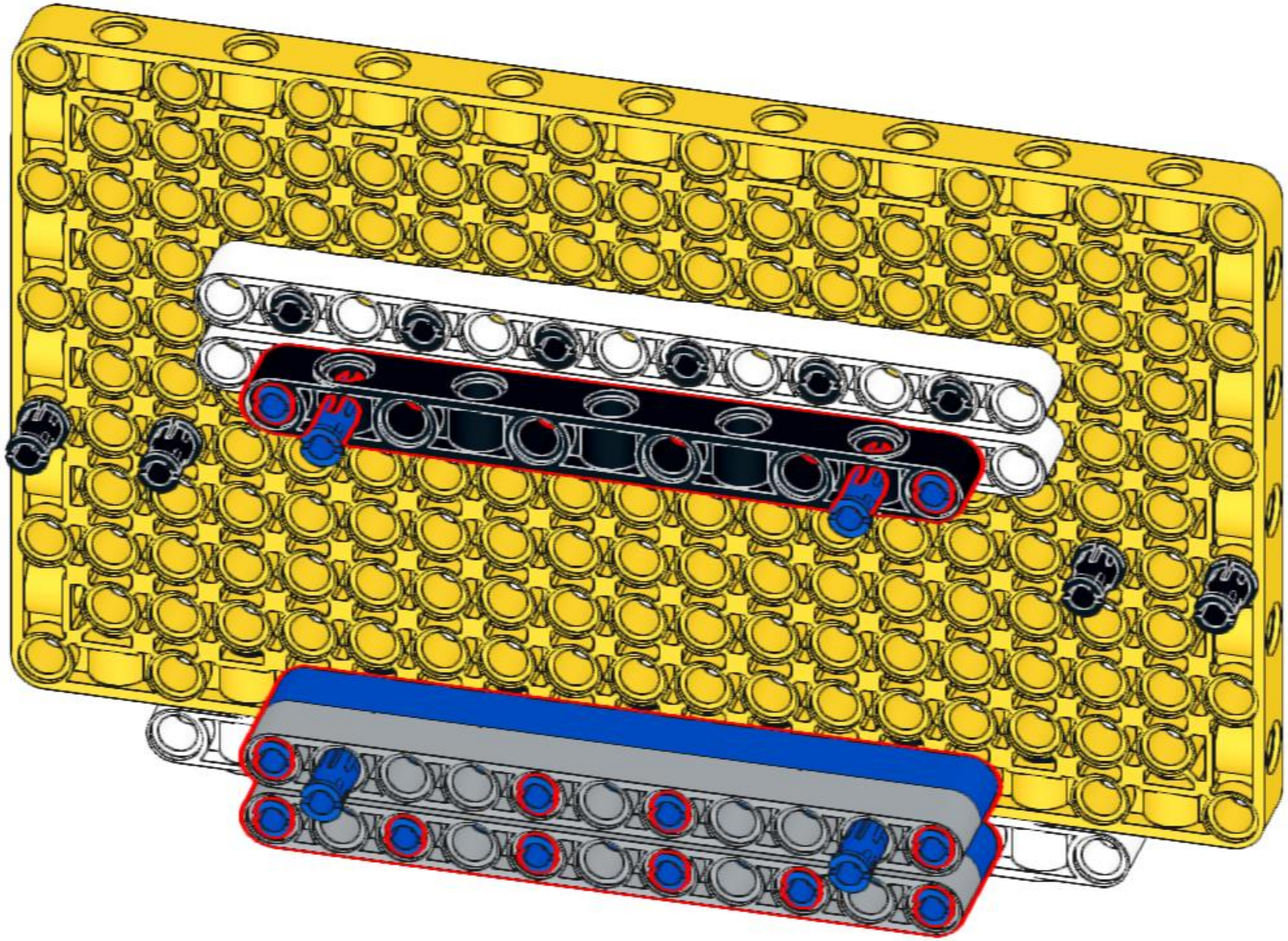

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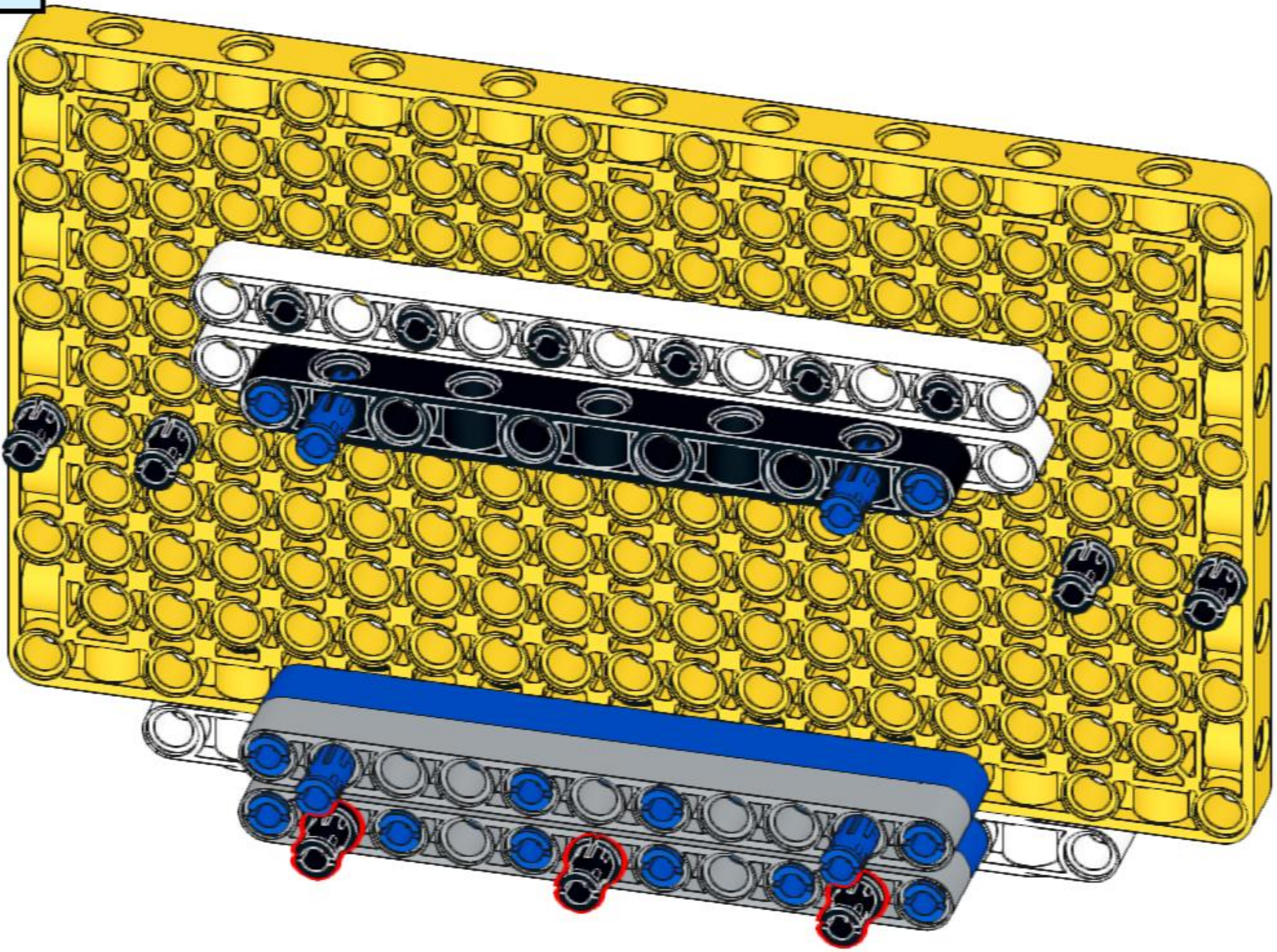

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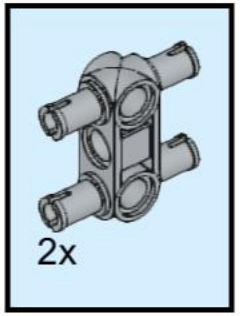

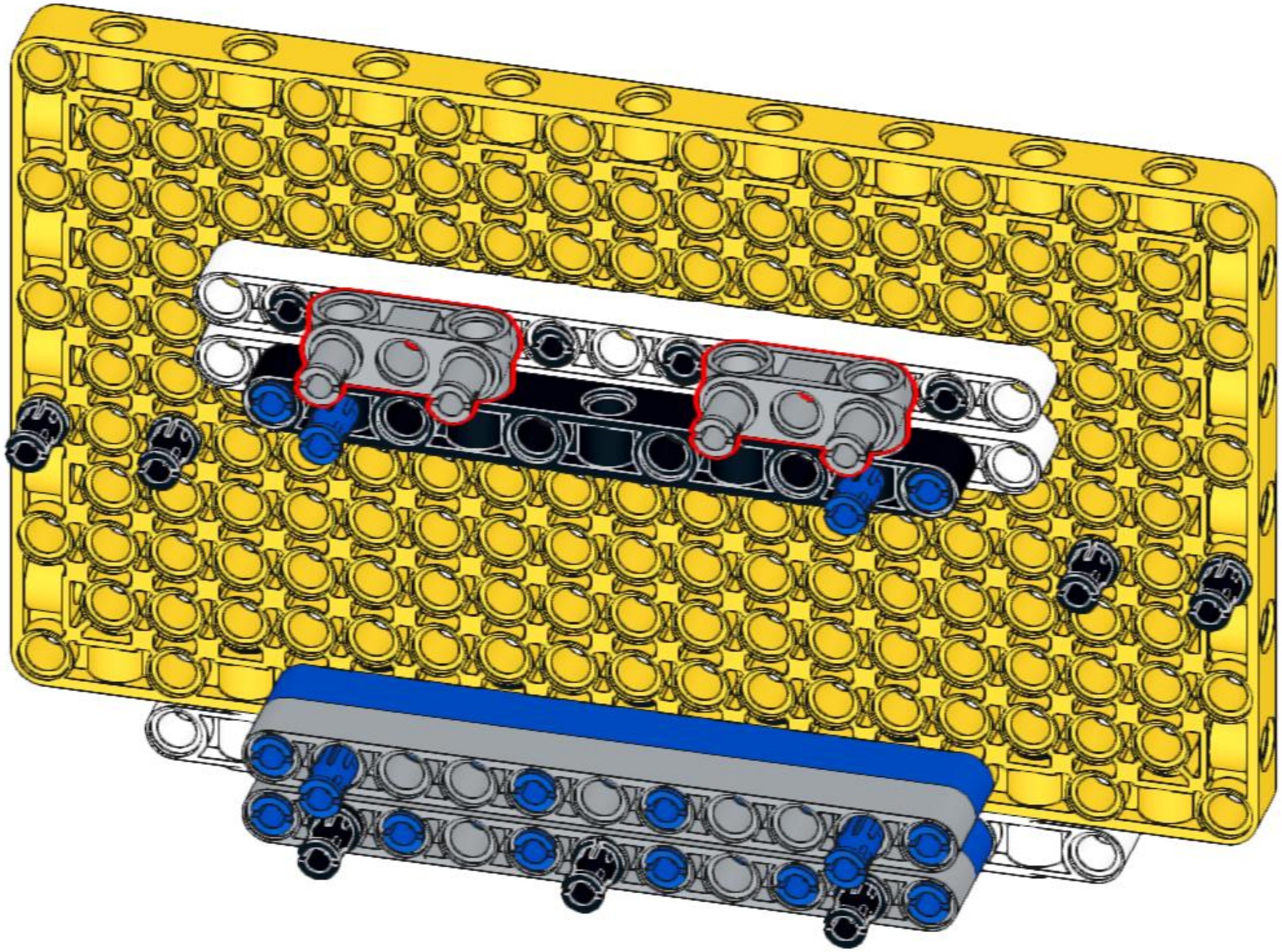

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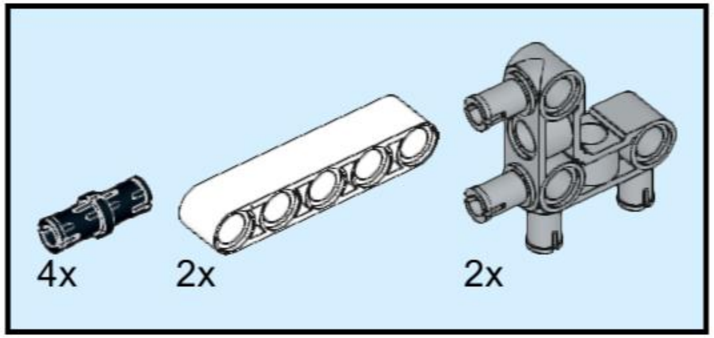

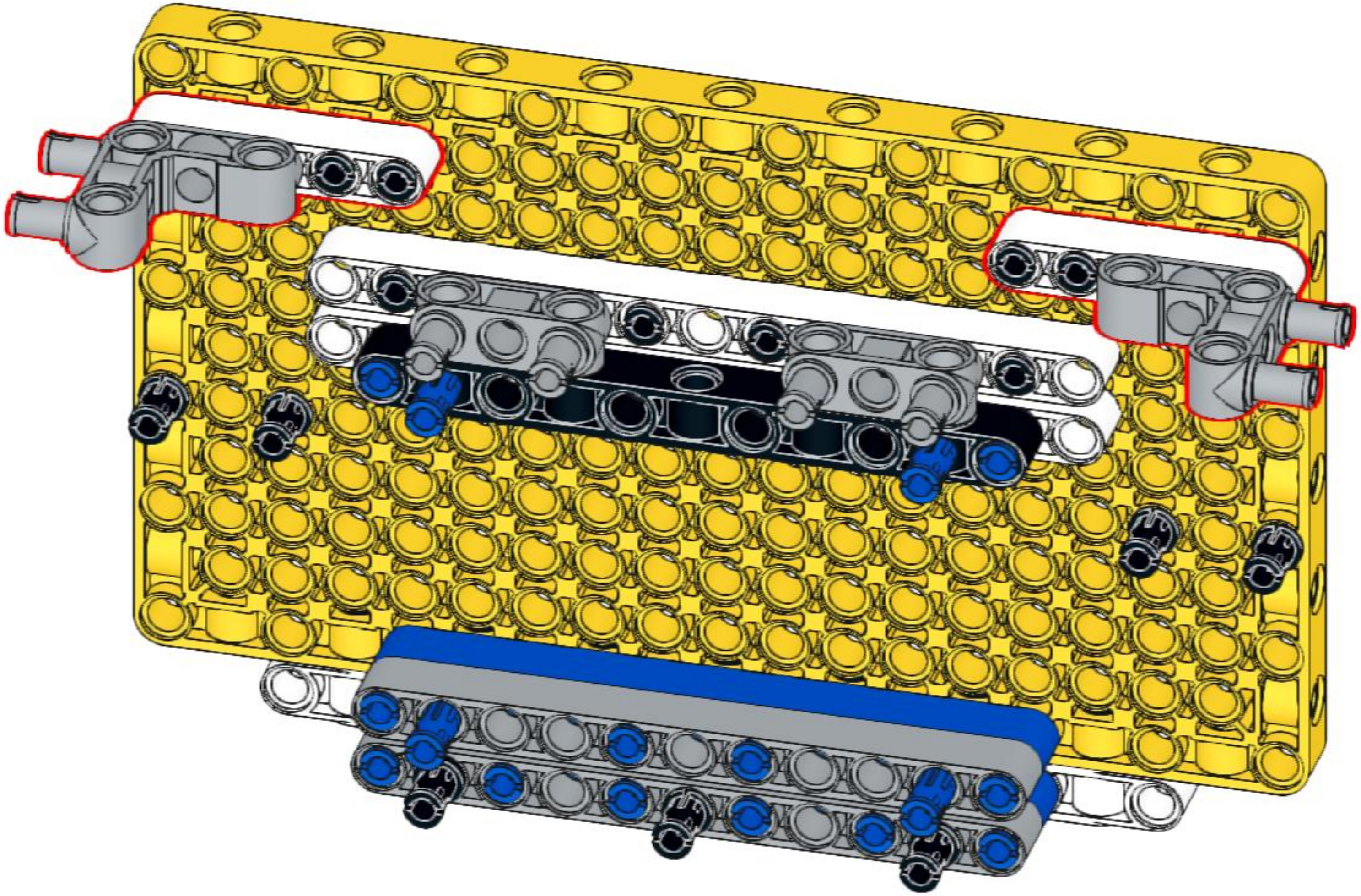

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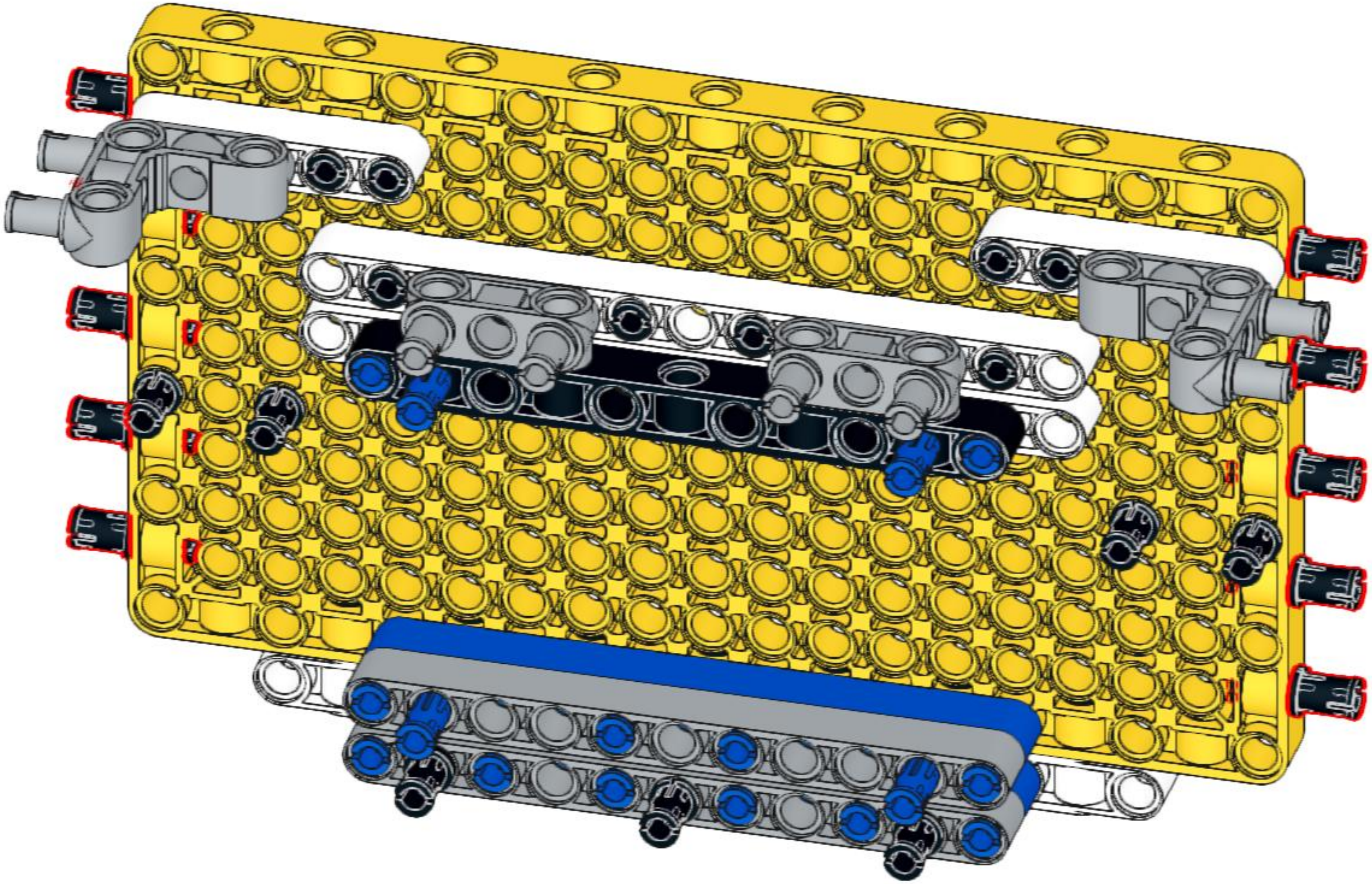

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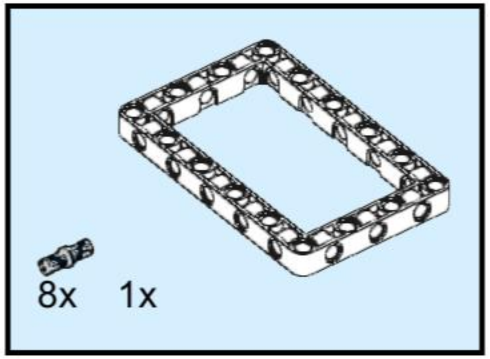

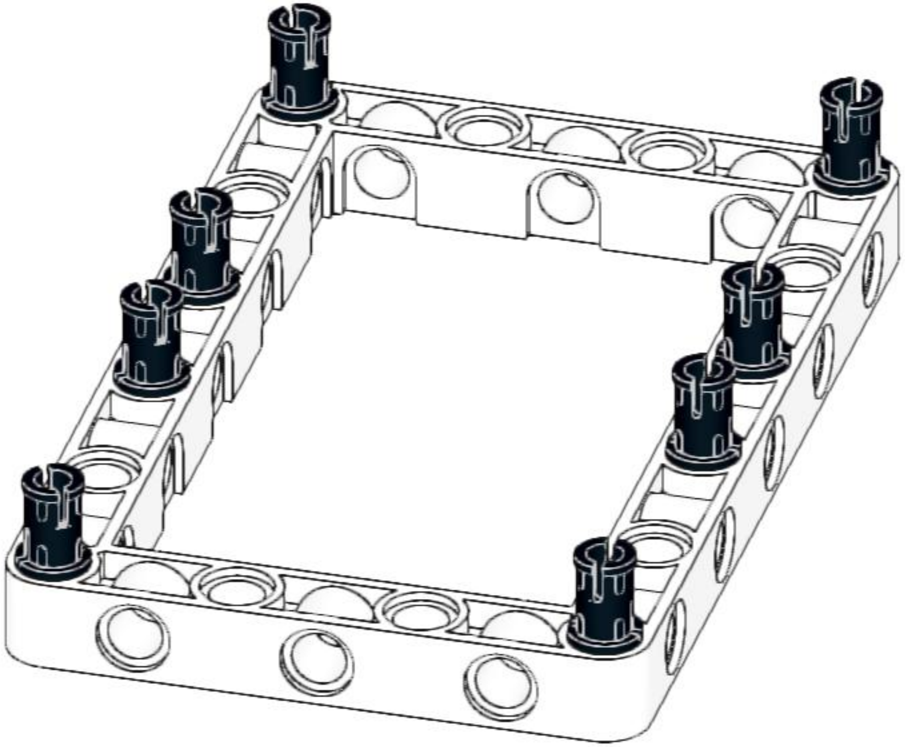

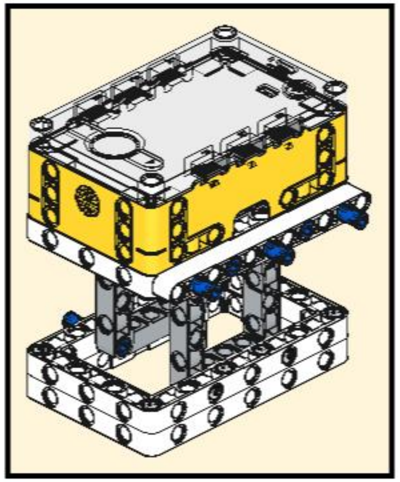

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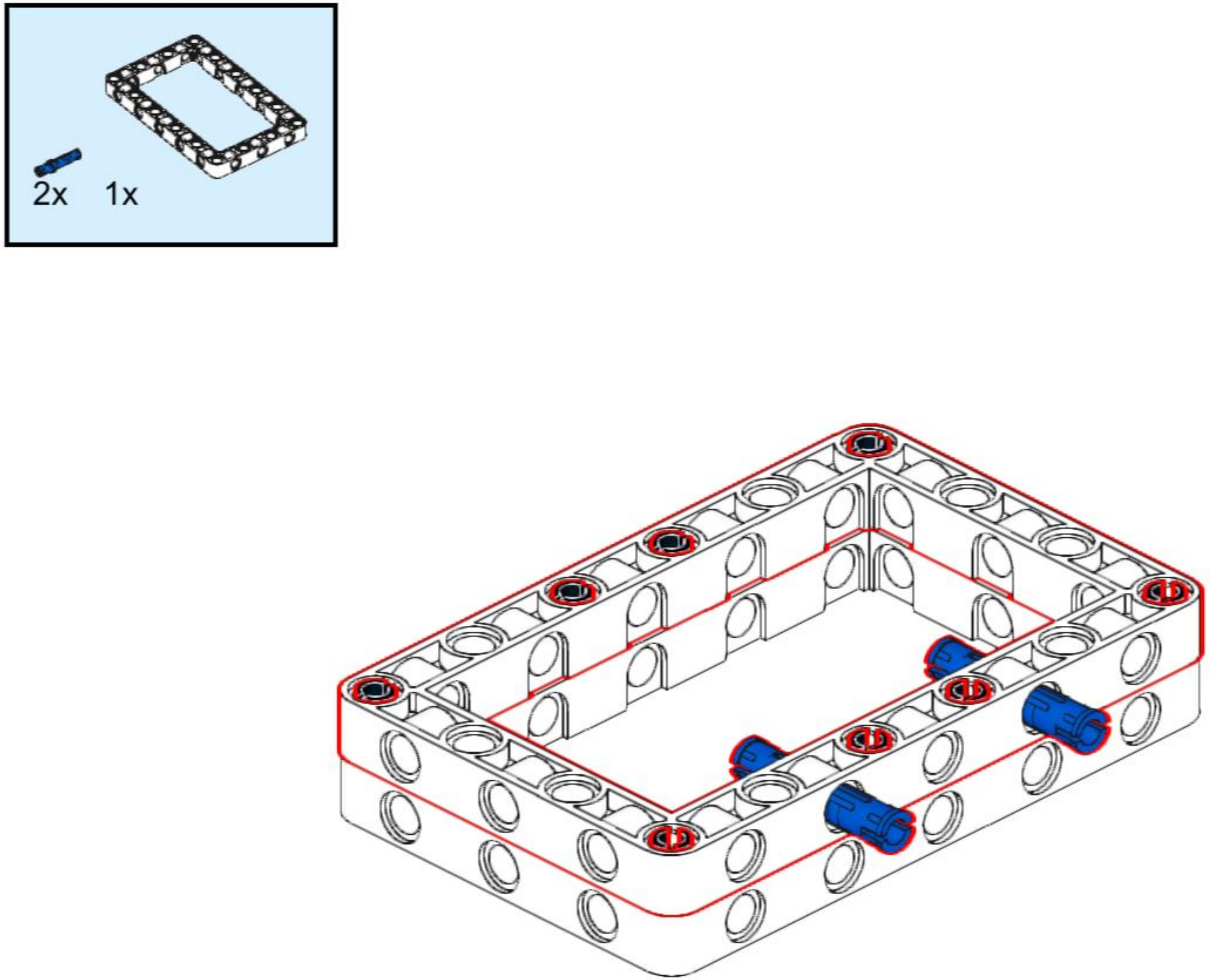

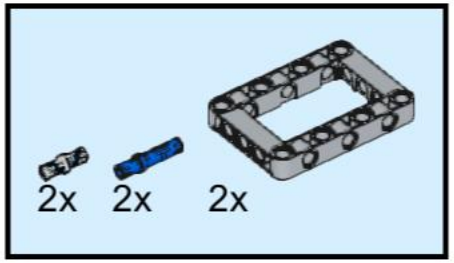

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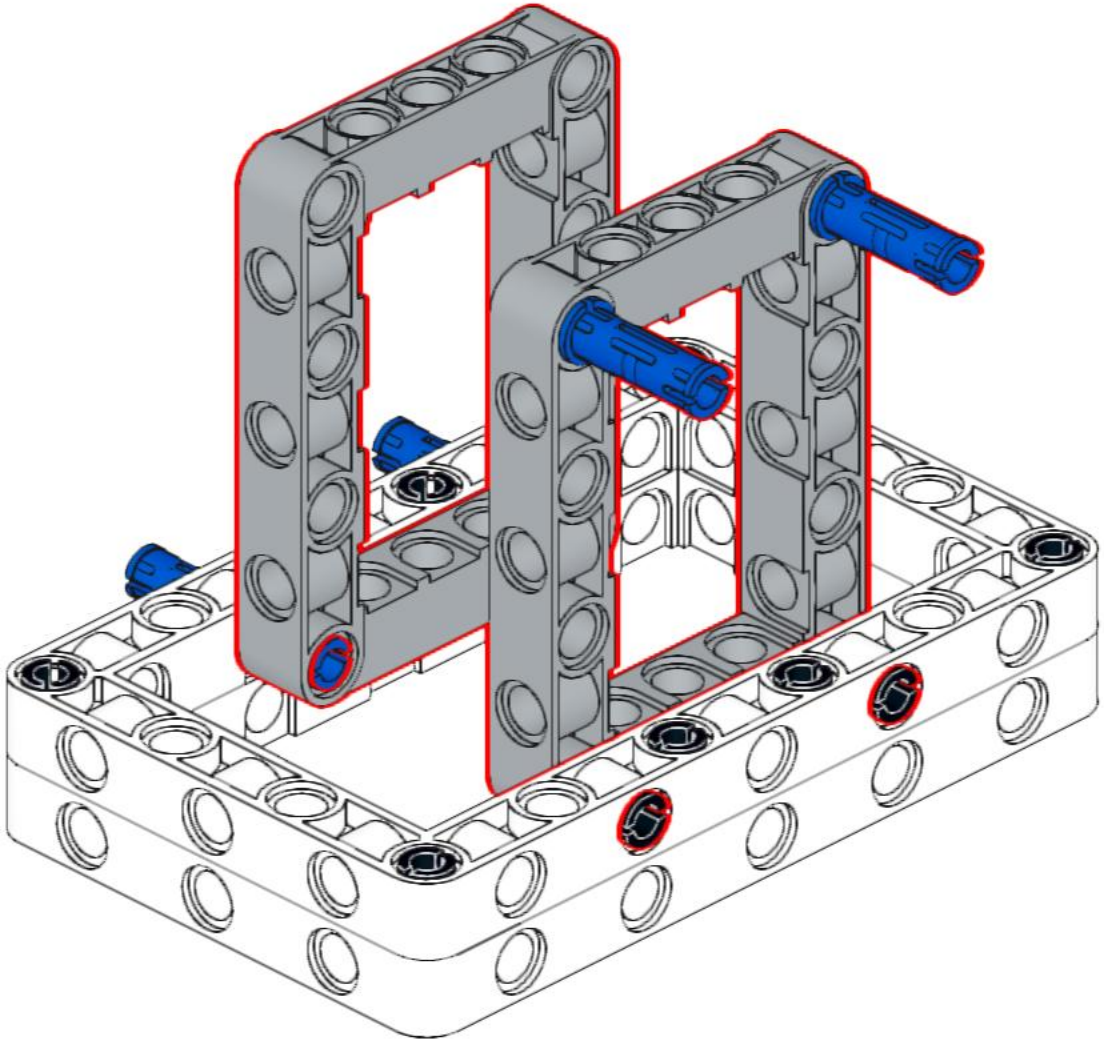

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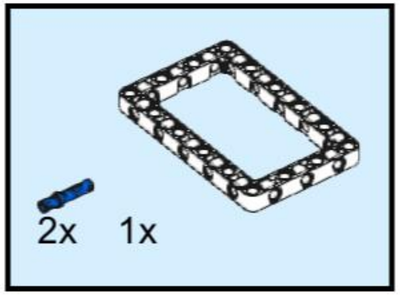

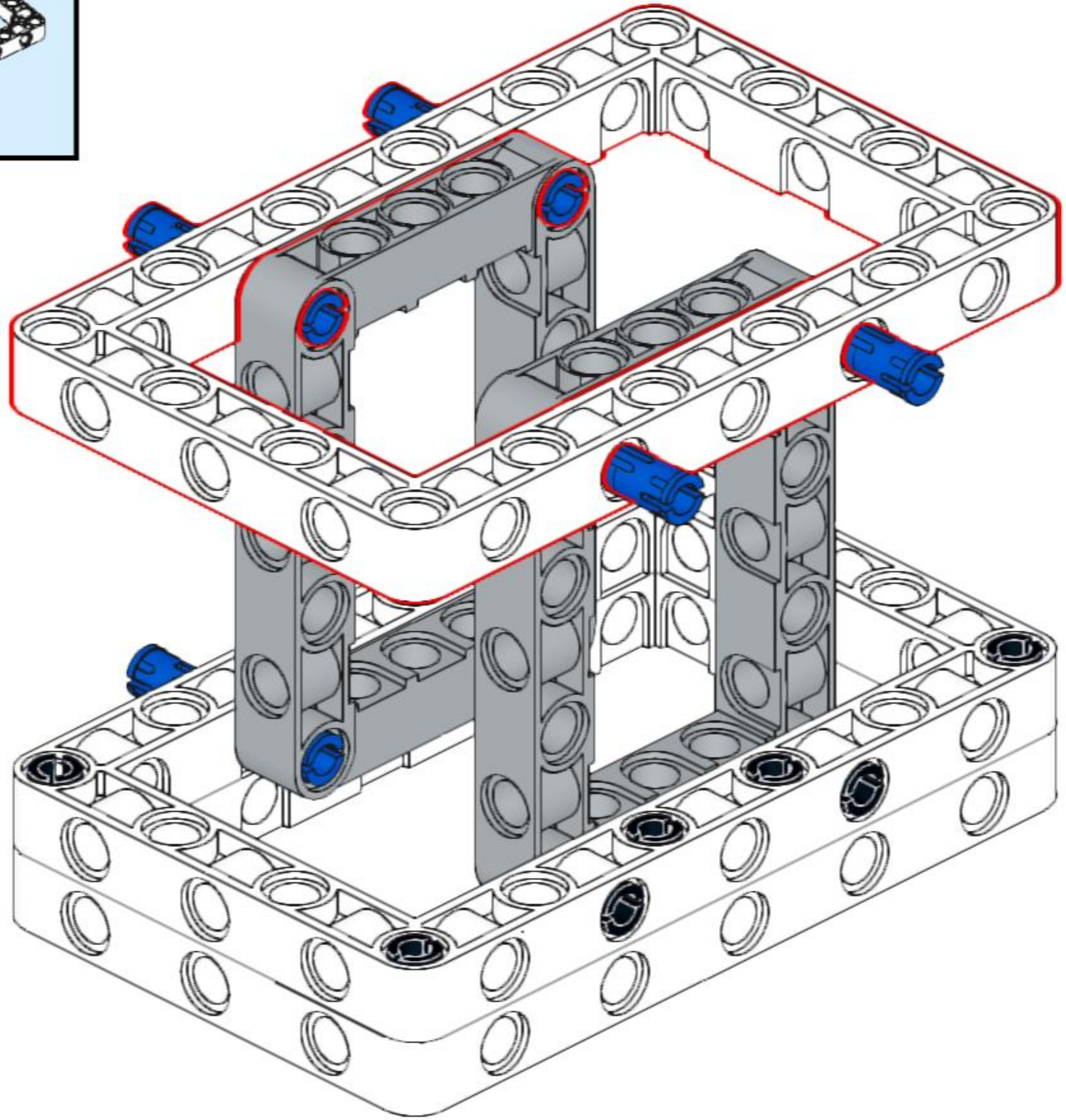

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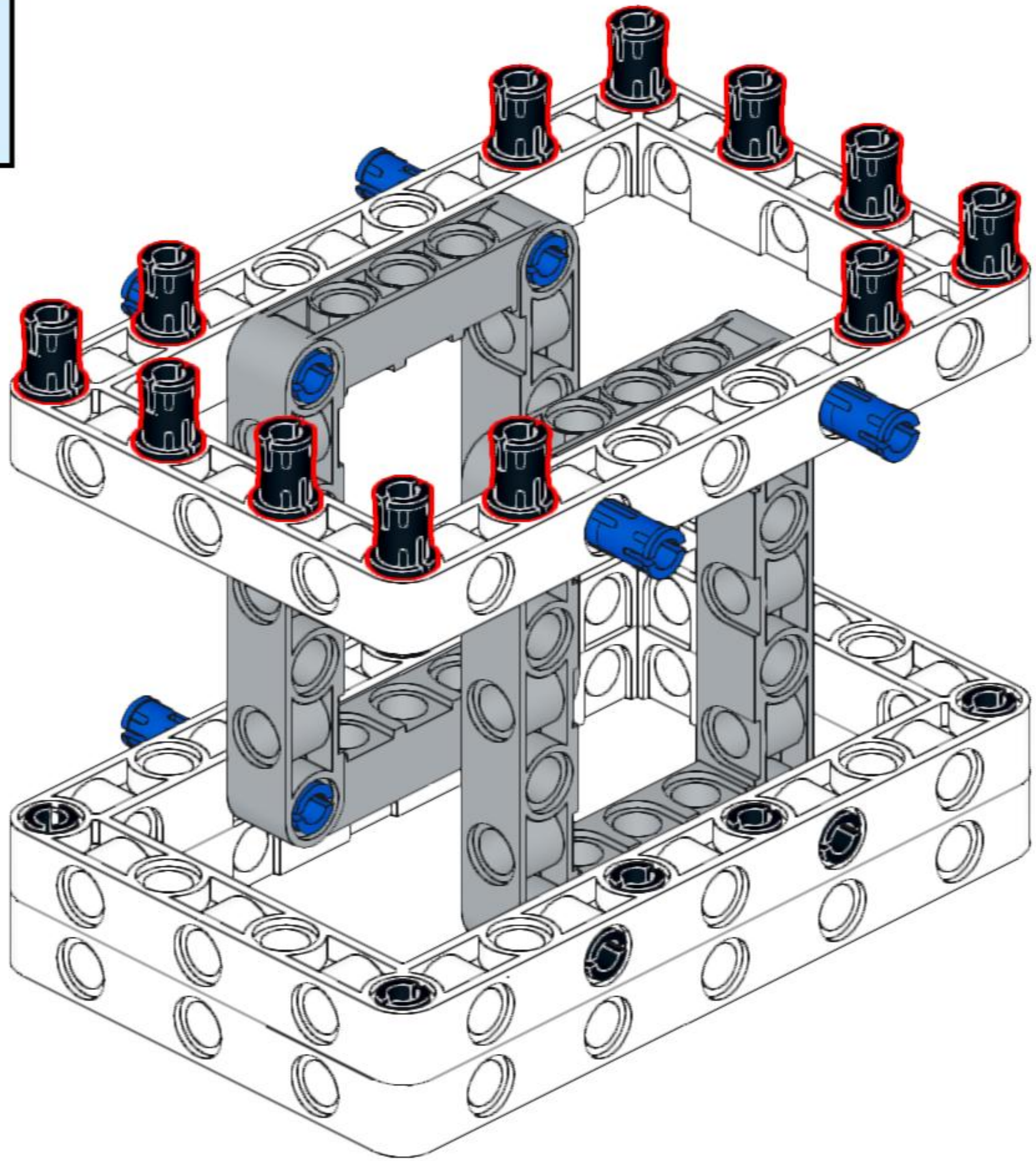

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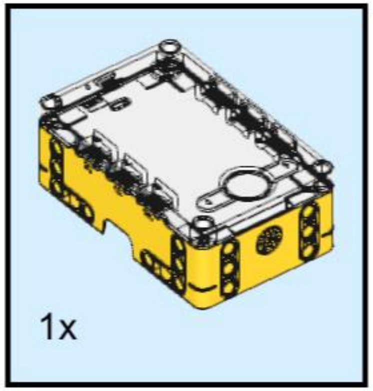

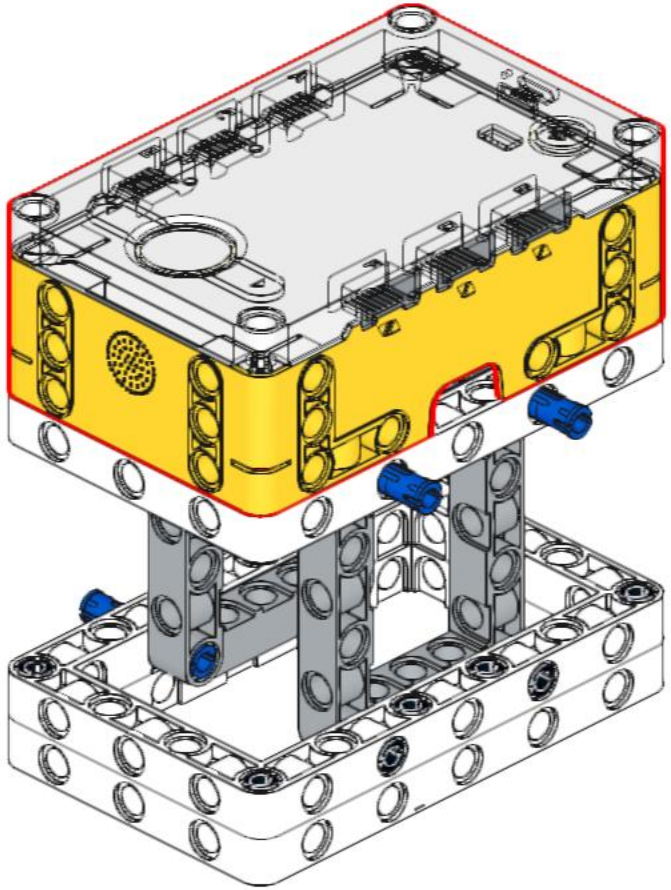

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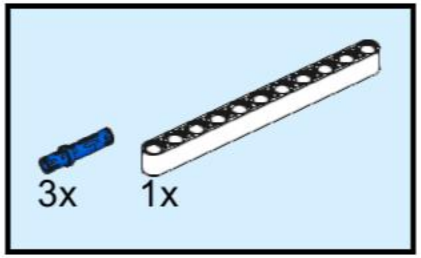

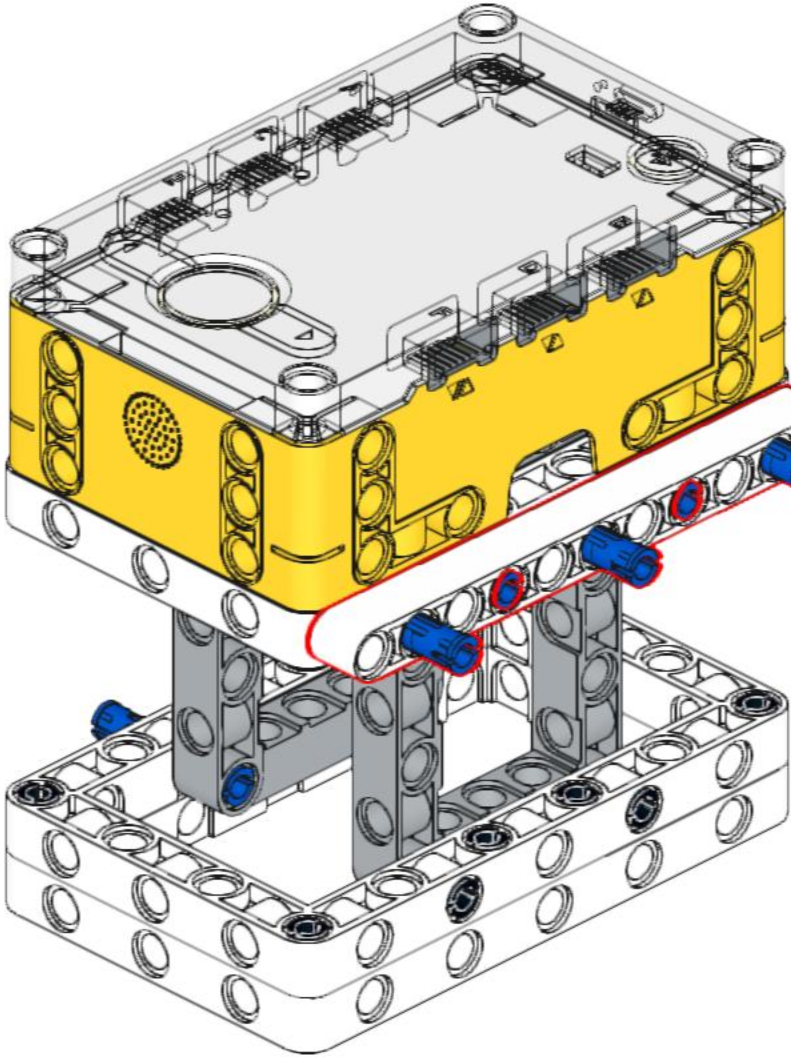

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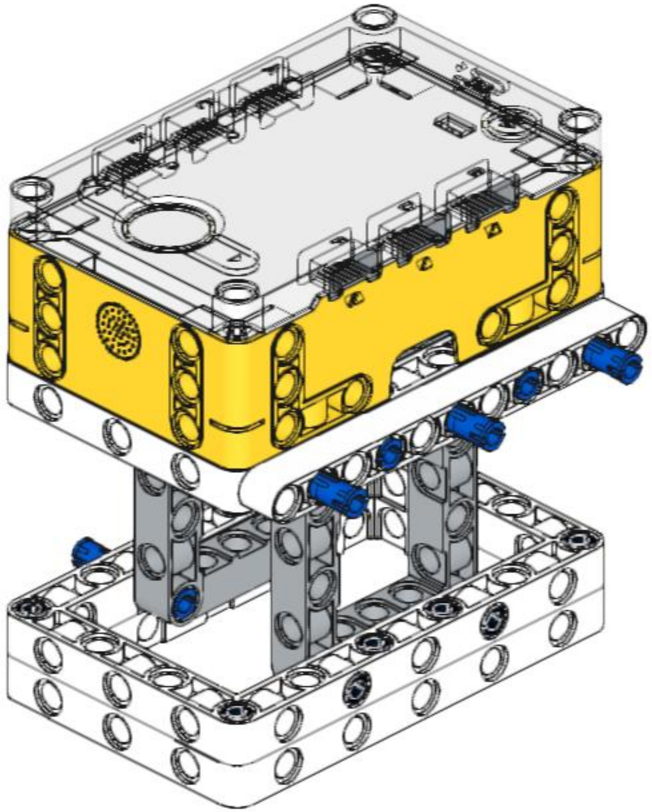

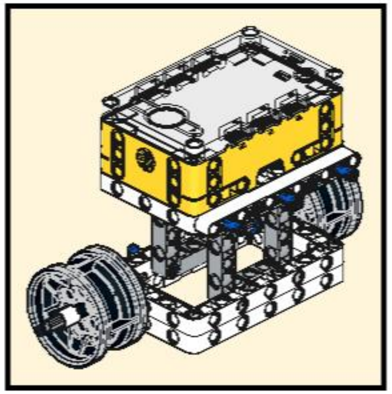

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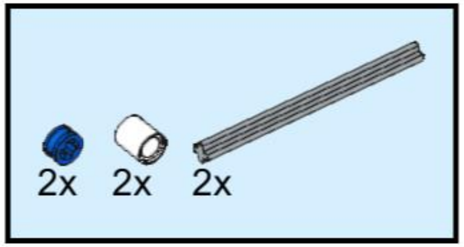

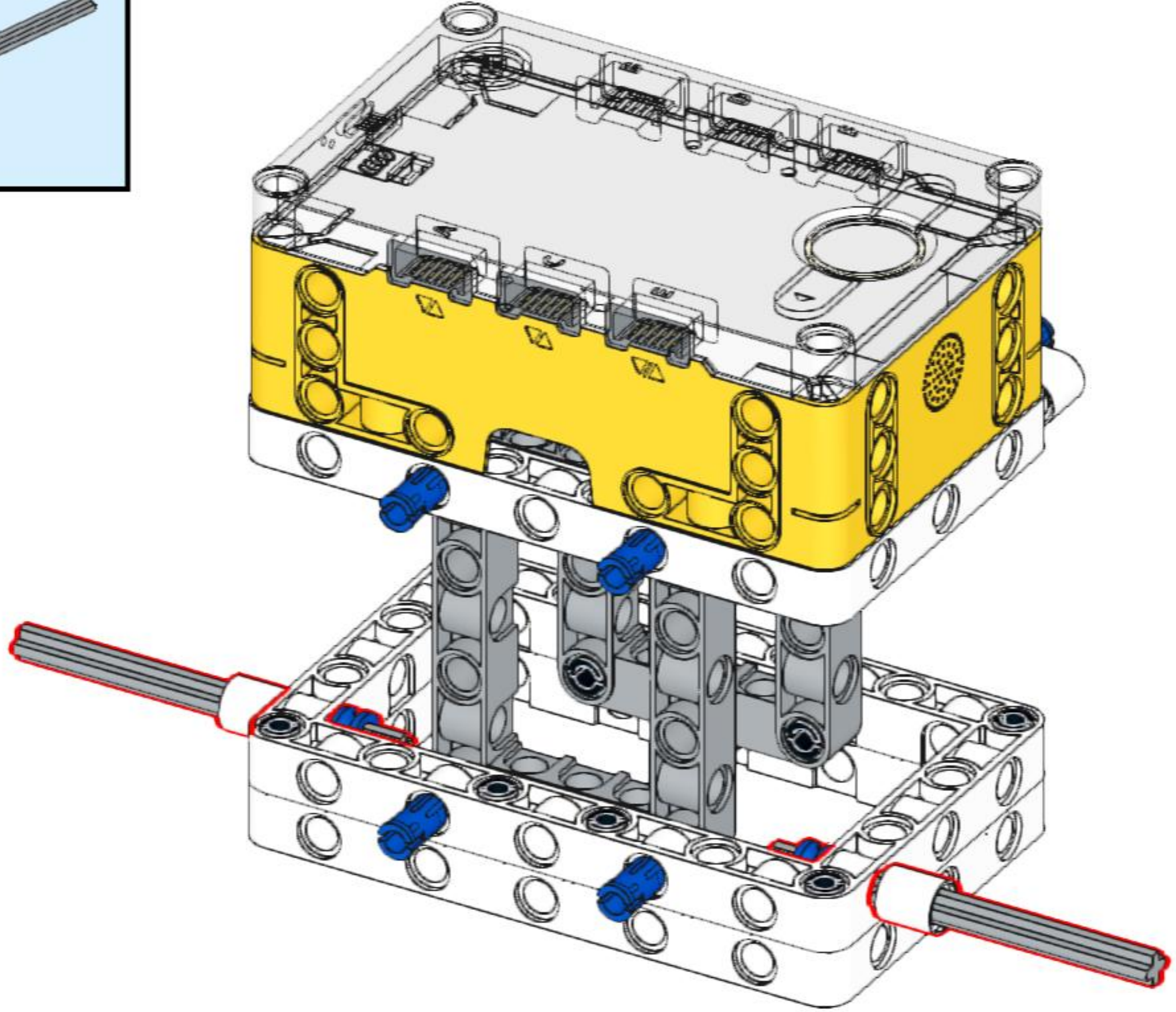

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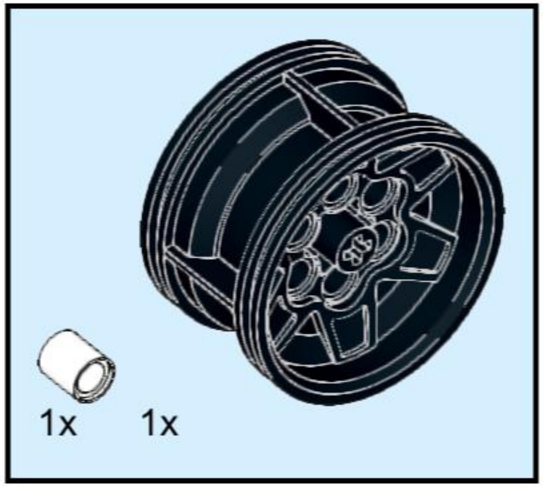

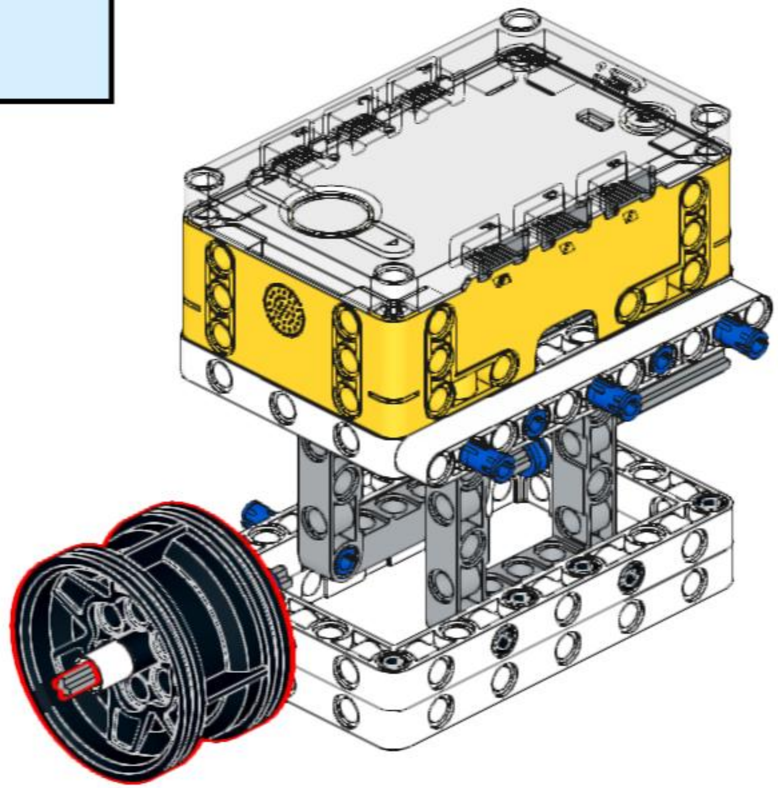

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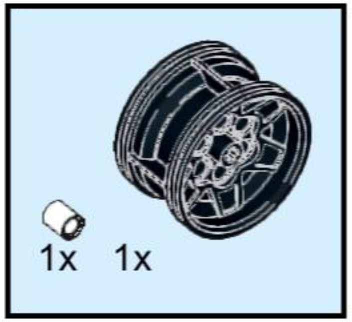

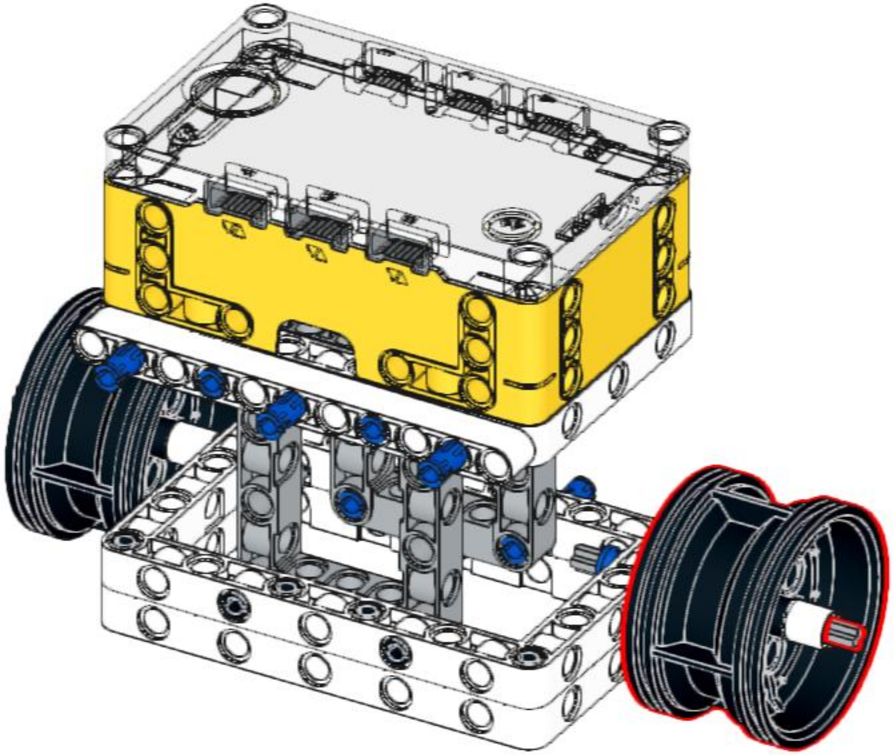

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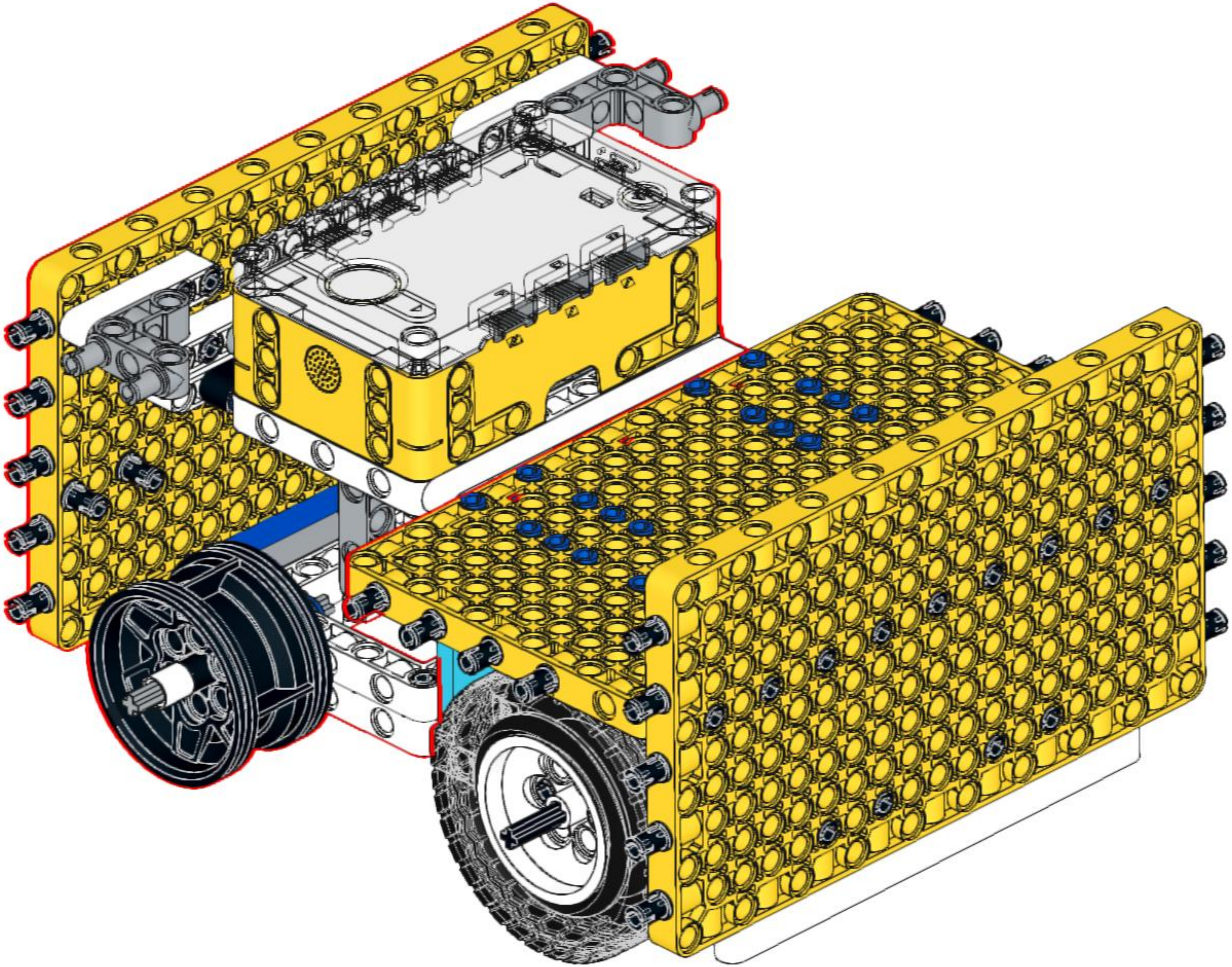

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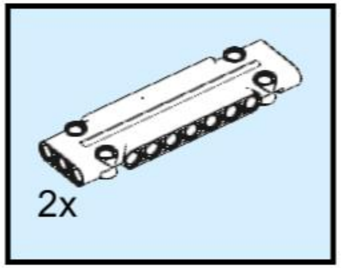

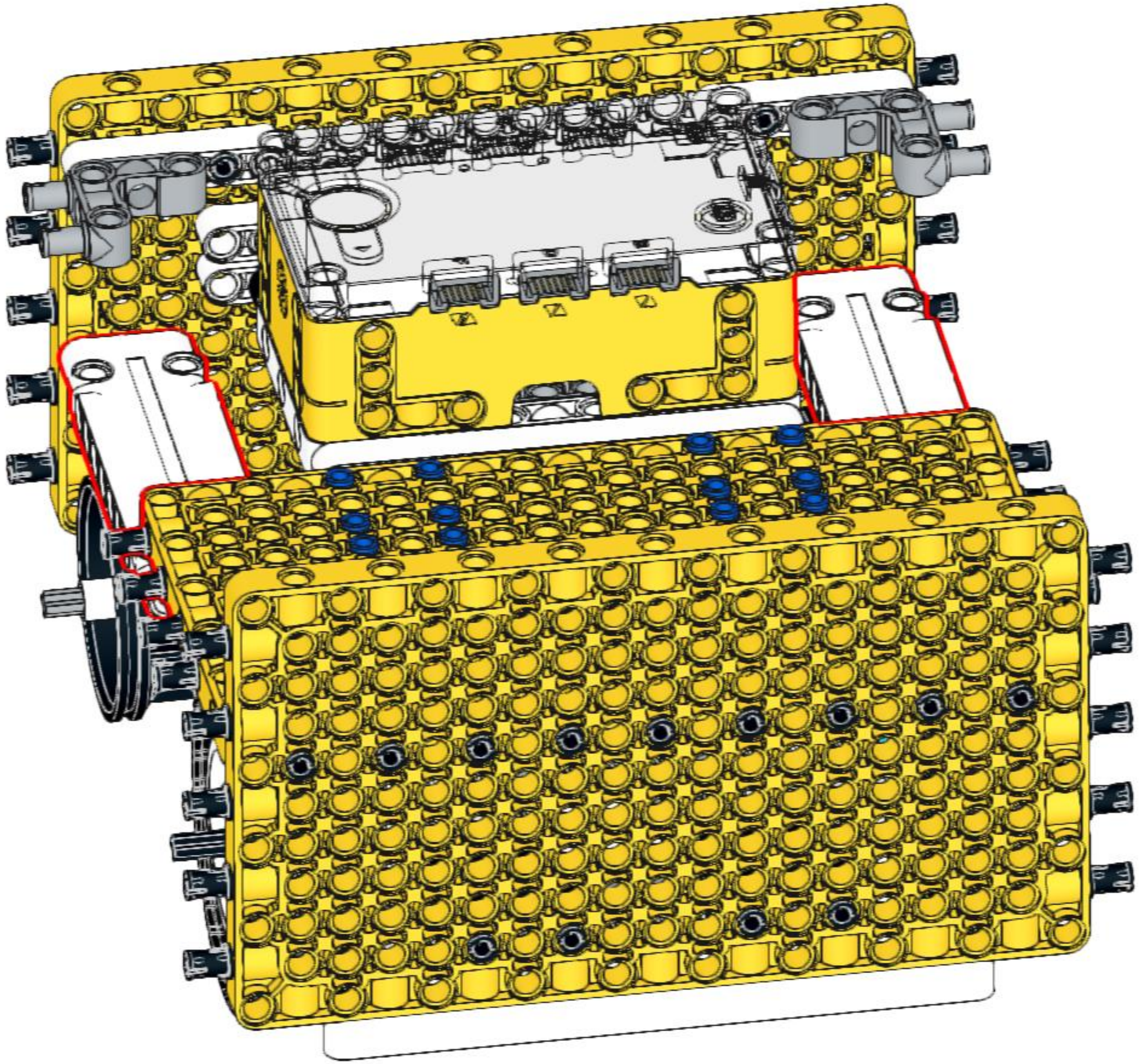

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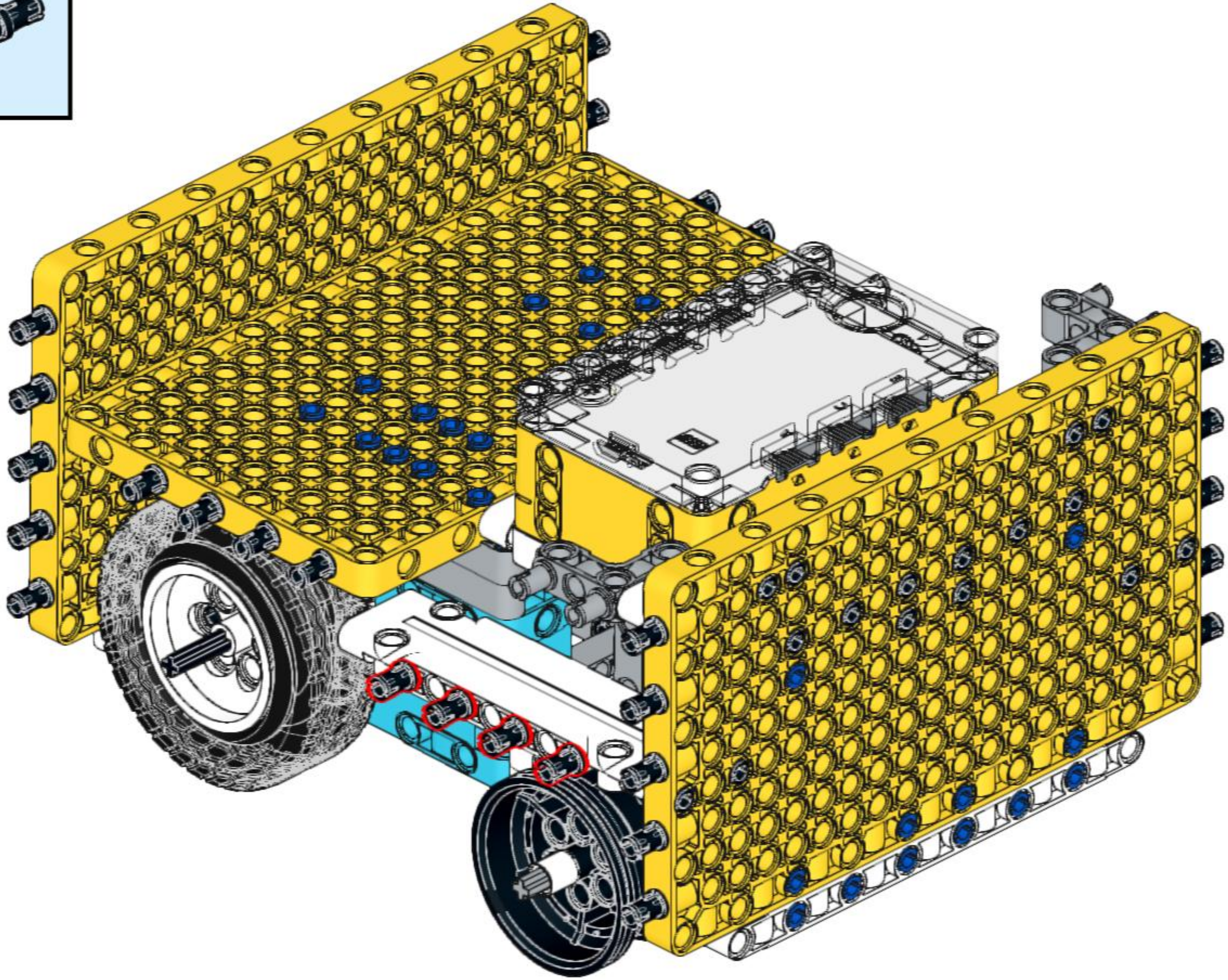

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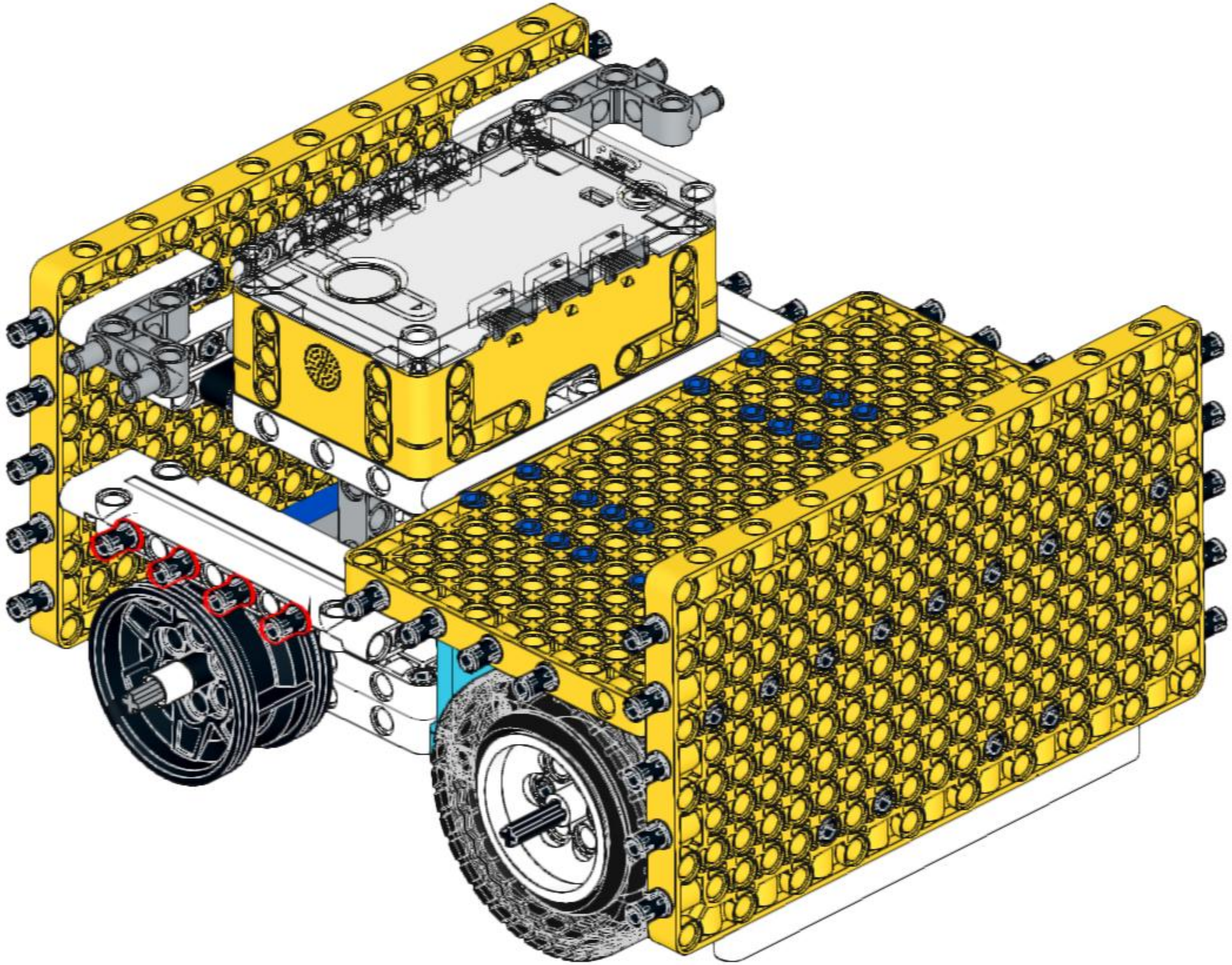

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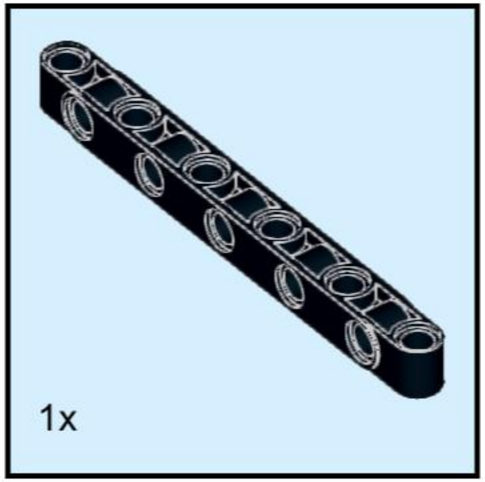

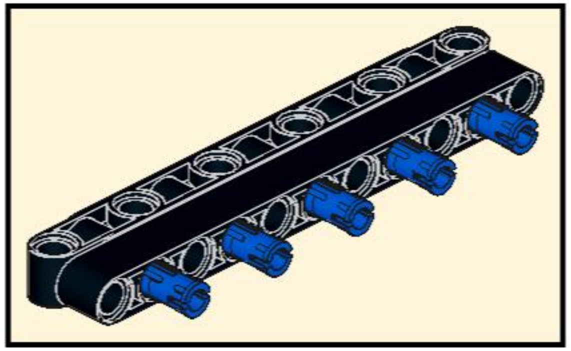

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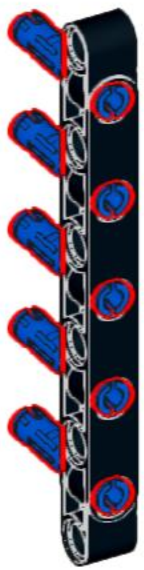

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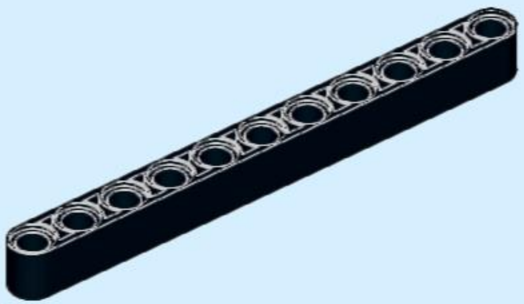

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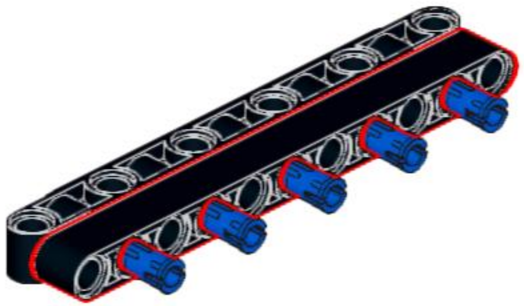

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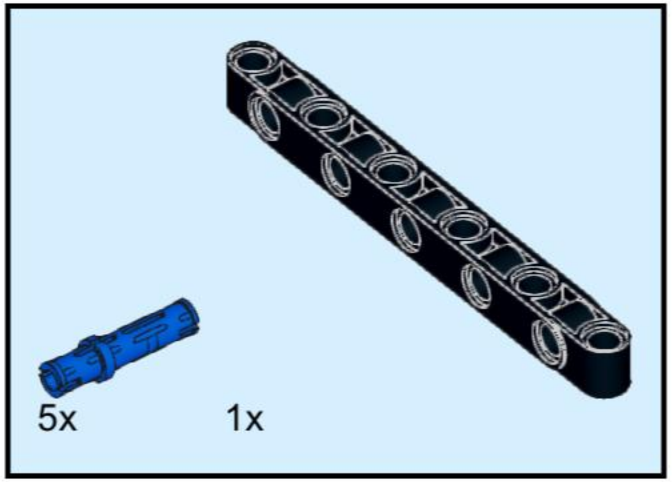

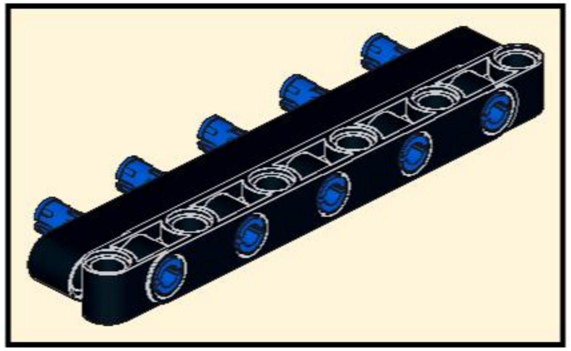

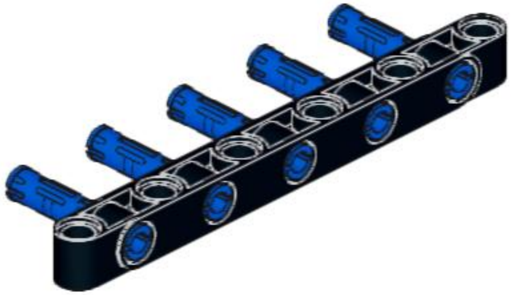

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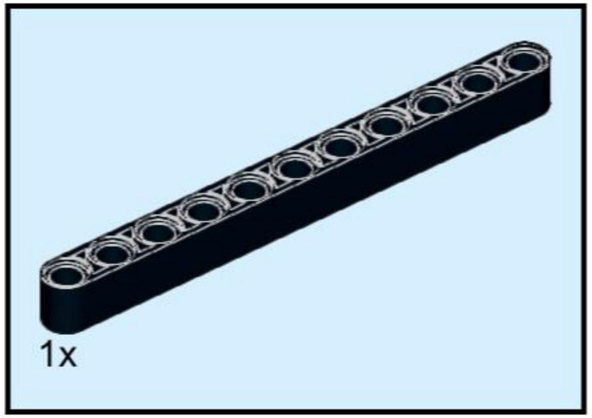

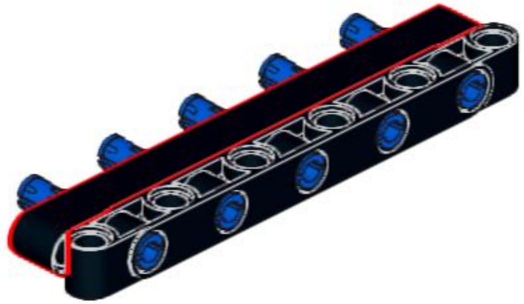

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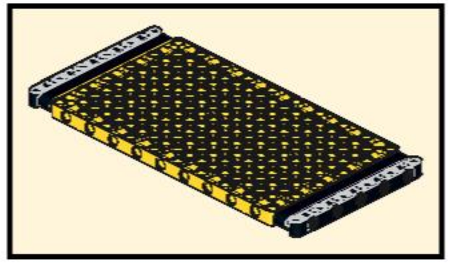

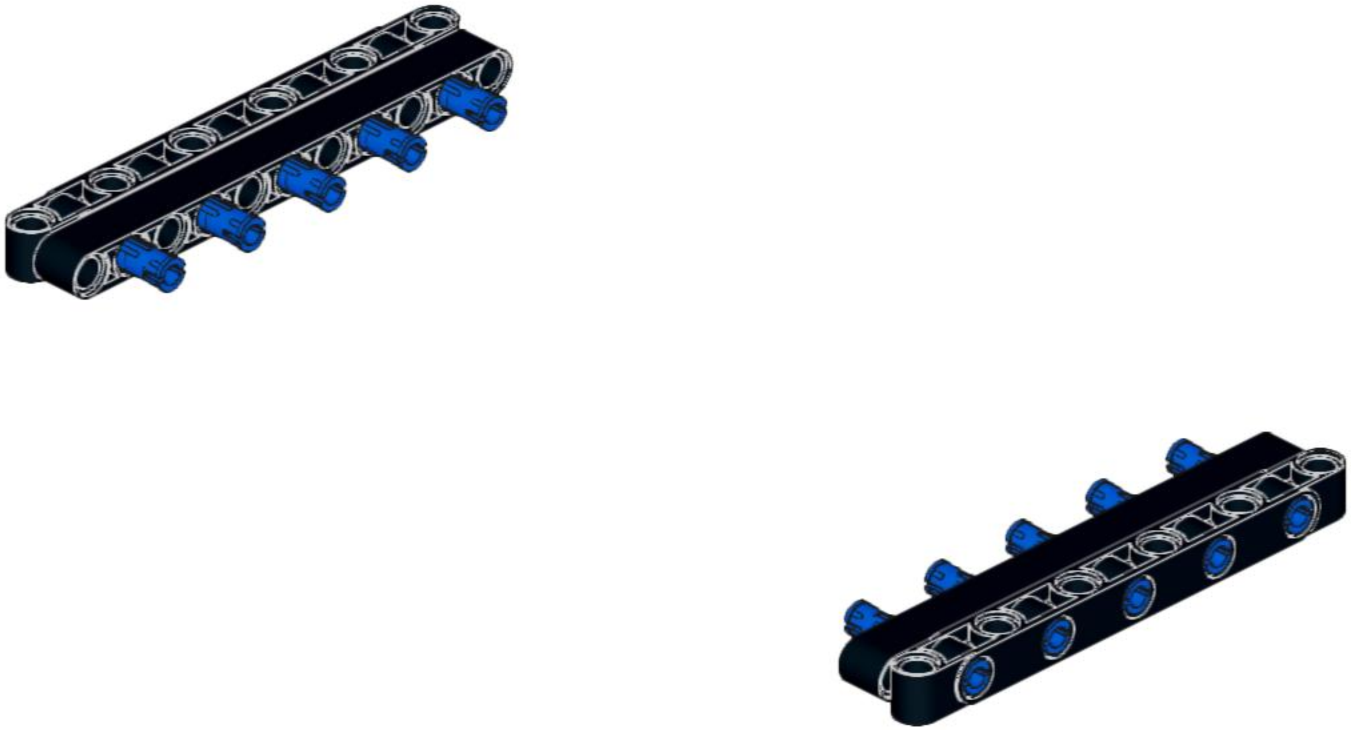

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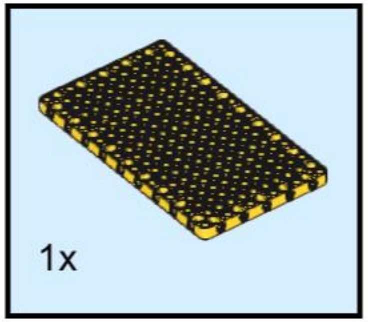

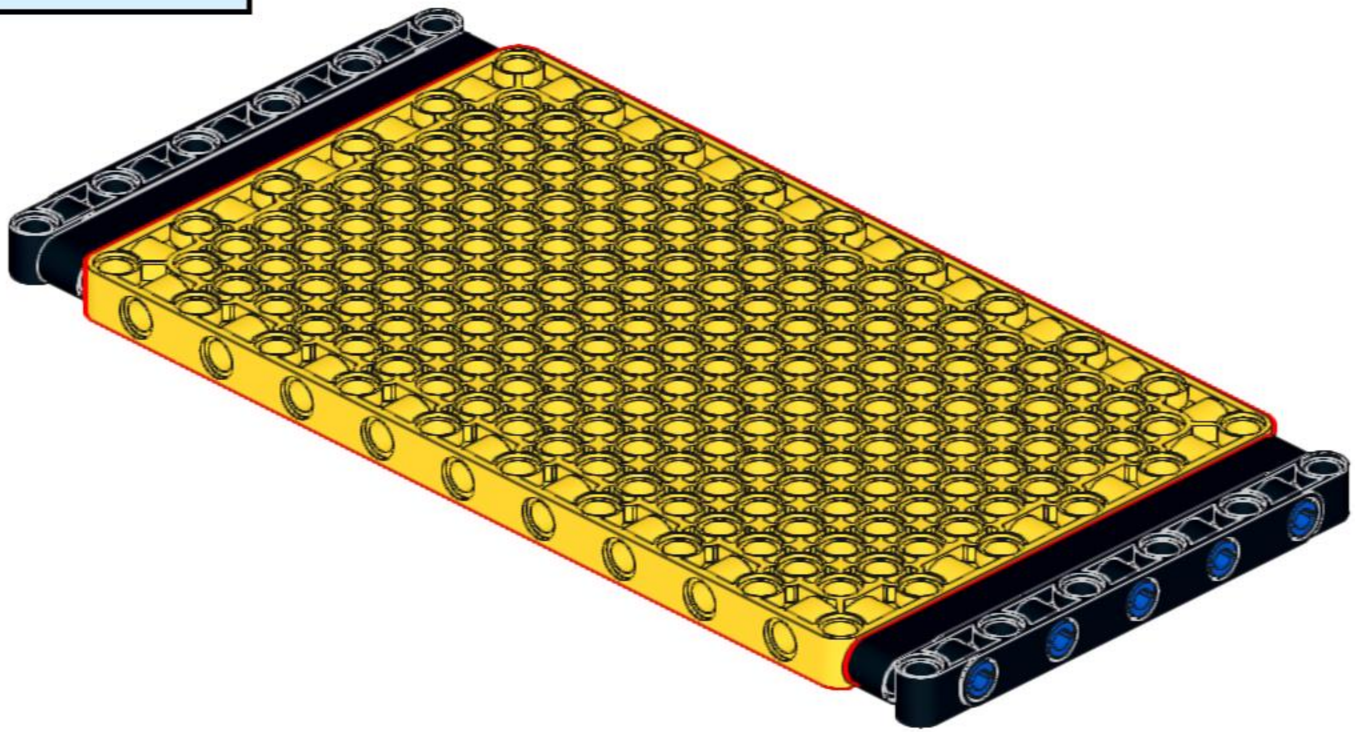

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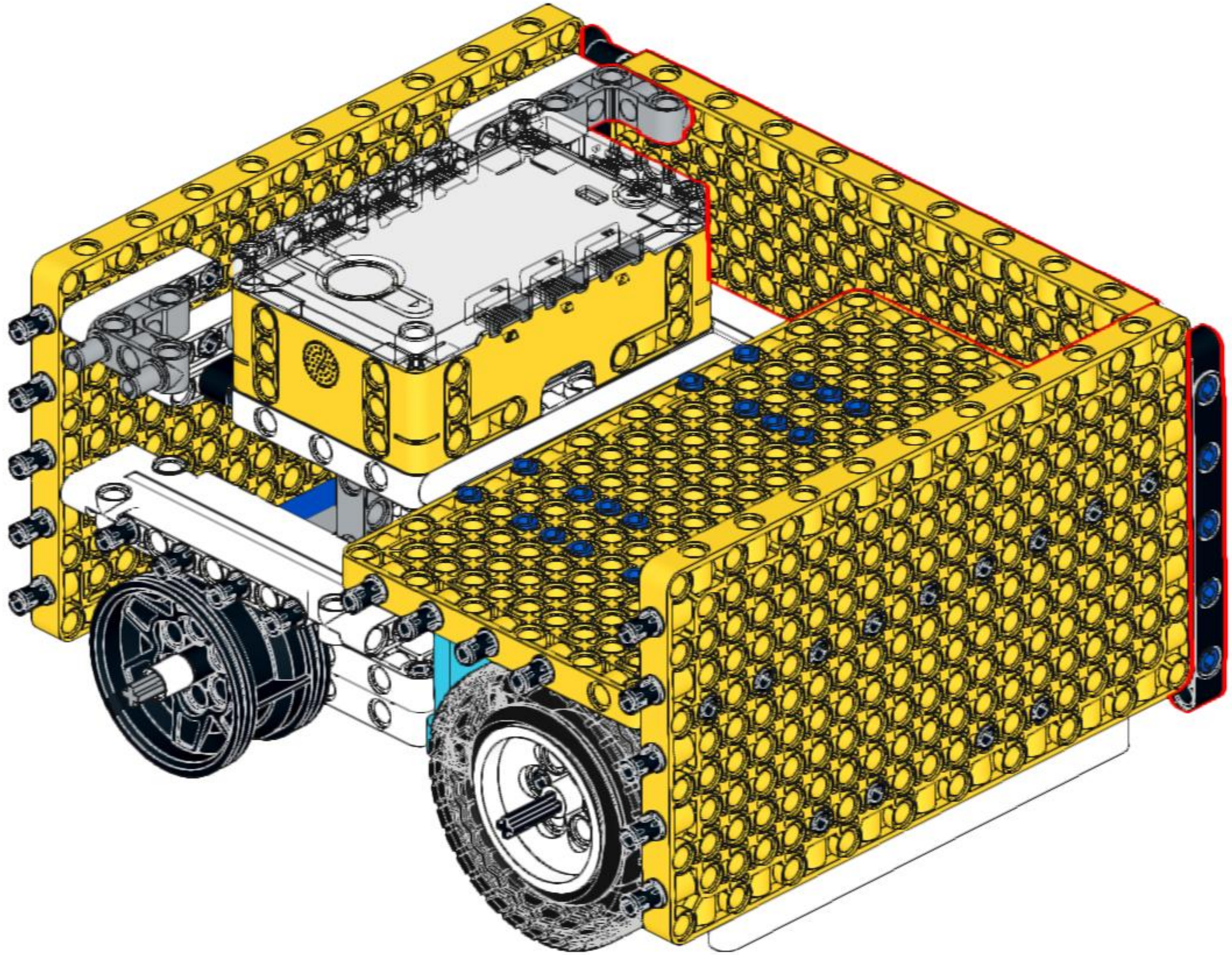

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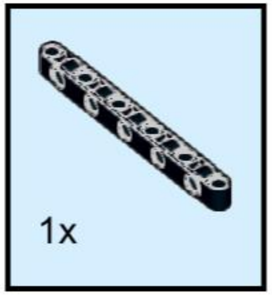

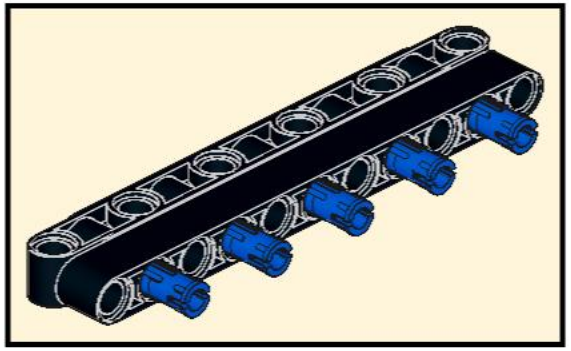

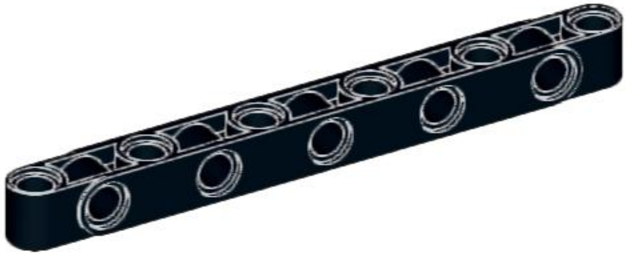

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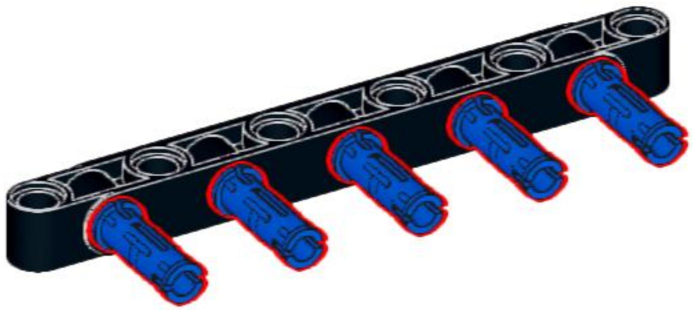

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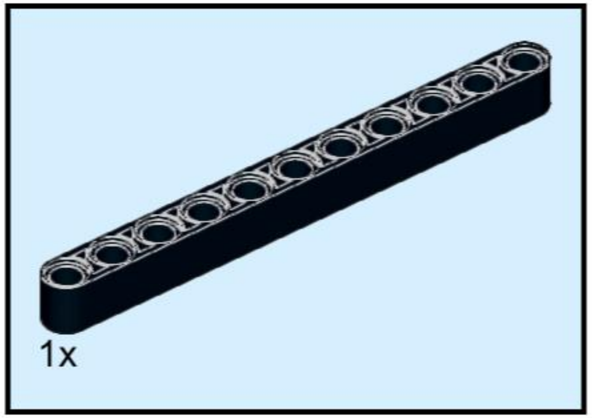

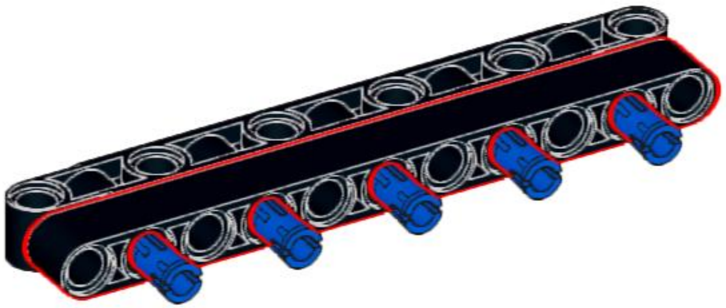

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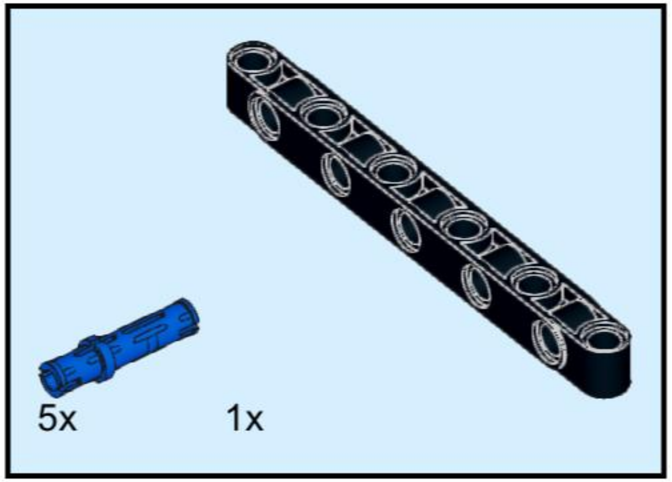

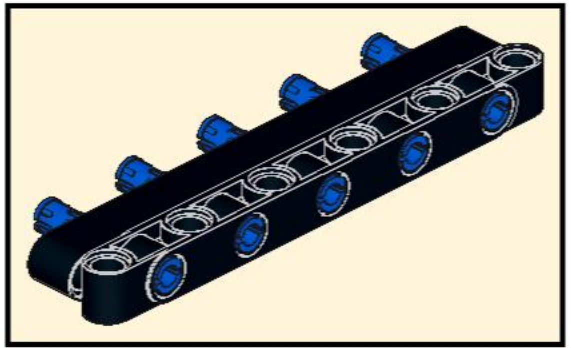

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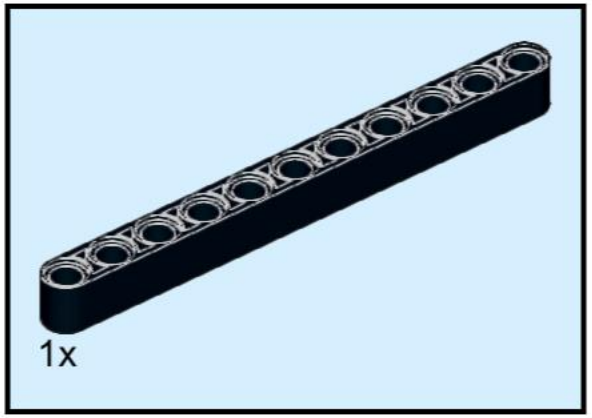

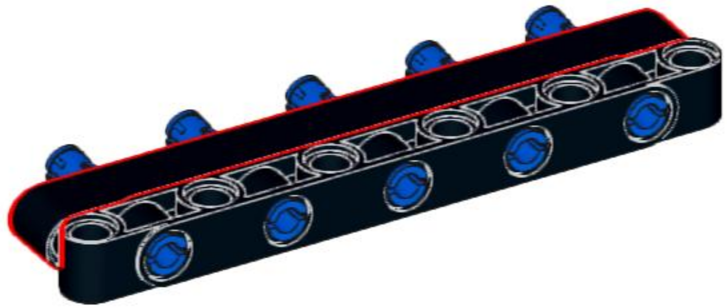

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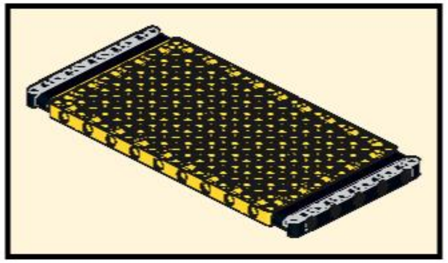

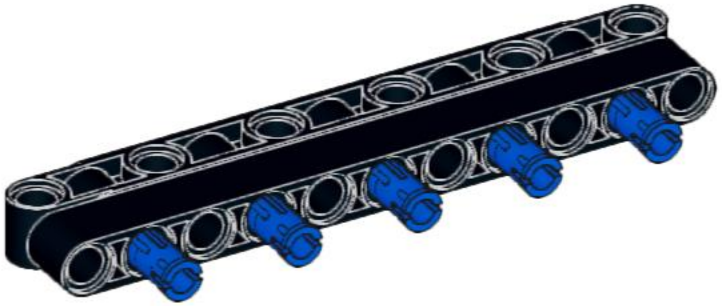

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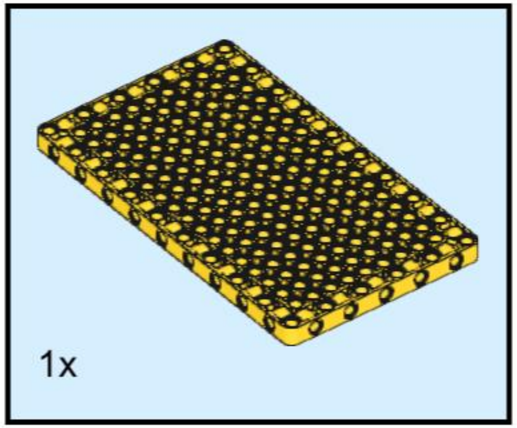

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